Introduction to Electrodynamics
Fourth Edition
David J. Griffiths
Solutions by David A. Lee

All errors, typographical and substantive, and other offenses, are entirely my own.

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1 - Vector Analysis

1.1 - Vector Algebra

1.1.1 - Vector Operations

Problem: 1.1

Using the definitions in Eqs. 1.1 and 1.4, and appropriate diagrams, show that the dot product and cross product are distributive,

(a) when the three vectors are coplanar;

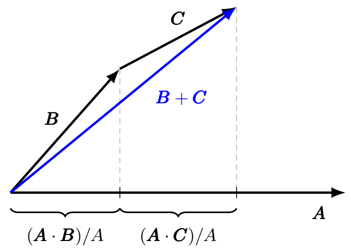
(b) in the general case.

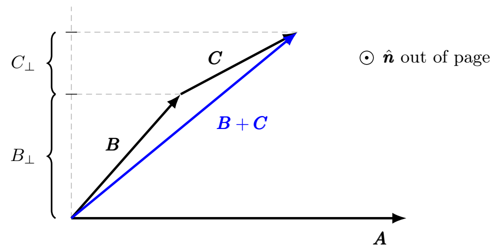
The dot and cross products are defined, respectively

$$\boldsymbol{A}\cdot\boldsymbol{B} \equiv AB\cos(\theta)\,, \qquad \boldsymbol{A}\times\boldsymbol{B} \equiv AB\sin(\theta)\,\hat{\boldsymbol{n}}$$

(a) We seek to demonstrate

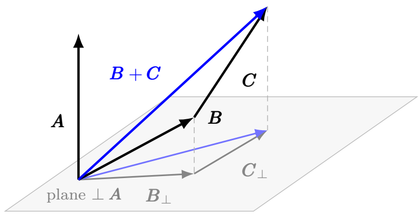
$$\boldsymbol{A}\cdot(\boldsymbol{B}+\boldsymbol{C}) = \boldsymbol{A}\cdot\boldsymbol{B}+\boldsymbol{A}\cdot\boldsymbol{C}, \qquad \boldsymbol{A}\times(\boldsymbol{B}+\boldsymbol{C}) = (\boldsymbol{A}\times\boldsymbol{B})+(\boldsymbol{A}\times\boldsymbol{C})$$

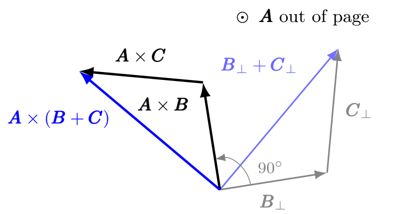




(b) In this part, distributivity looks like

$$\boldsymbol{A}\times(\boldsymbol{B}+\boldsymbol{C}) = (\boldsymbol{A}\times\boldsymbol{B})+(\boldsymbol{A}\times\boldsymbol{C})$$





rotating each shadow 90° about $\hat{\boldsymbol{A}}$ and scaling by A gives the cross products (lengths not to scale)

Problem: 1.2

Is the cross product associative?

$$(\boldsymbol{A}\times\boldsymbol{B})\times\boldsymbol{C} \stackrel{?}{=} \boldsymbol{A}\times(\boldsymbol{B}\times\boldsymbol{C})$$

If so, *prove* it; if not, provide a counterexample (the simpler the better).

No. Consider the case where $\boldsymbol{A}=\boldsymbol{B}\neq\boldsymbol{0}$ and $\boldsymbol{C}\neq\boldsymbol{0}$ and $\boldsymbol{C}\nparallel\boldsymbol{A}$. Then the left-hand side is the zero vector:

$$(\boldsymbol{A}\times\boldsymbol{A})\times\boldsymbol{C}=\boldsymbol{0}$$

but not on the right as $\boldsymbol{A}\times\boldsymbol{C}\neq\boldsymbol{0}$ and since $\boldsymbol{C}\nparallel\boldsymbol{A}$, we have $\boldsymbol{A}\nparallel(\boldsymbol{A}\times\boldsymbol{C})$. Thus

$$\boldsymbol{A}\times(\boldsymbol{A}\times\boldsymbol{C})\neq\boldsymbol{0}$$

1.1.2 - Vector Algebra: Component Form

Problem: 1.3

Find the angle between the body diagonals of a cube.

Let the cube have $l=w=h=a$. Situating both body diagonals at the origin, we have

$$\begin{aligned}\boldsymbol{d_1}&=a\hat{\boldsymbol{x}}+a\hat{\boldsymbol{y}}+a\hat{\boldsymbol{z}}\\ \boldsymbol{d_2}&=-a\hat{\boldsymbol{x}}+a\hat{\boldsymbol{y}}+a\hat{\boldsymbol{z}}\end{aligned}$$

Both of which have the same magnitude $\sqrt{3}a$. Computing component and projection forms, we have

$$\begin{aligned}\boldsymbol{d_1}\cdot\boldsymbol{d_2}&=a^2\\ &=d_1d_2\cos(\theta)\\ &=3a^2\cos(\theta)\end{aligned}$$

implying that $\theta=\arccos(1/3)\approx70.5°$.

Problem: 1.4

Use the cross product to find the components of the unit vector $\hat{\boldsymbol{n}}$ perpendicular to the shaded plane in Fig. 1.11.

One approach is to define

$$\begin{aligned}\boldsymbol{A}&=0\hat{\boldsymbol{x}}-2\hat{\boldsymbol{y}}+3\hat{\boldsymbol{z}}\\ \boldsymbol{B}&=1\hat{\boldsymbol{x}}-2\hat{\boldsymbol{y}}+0\hat{\boldsymbol{z}}\end{aligned}$$

Then the cross product is calculated as

$$\boldsymbol{A}\times\boldsymbol{B}=\det\begin{pmatrix}\hat{\boldsymbol{x}}&\hat{\boldsymbol{y}}&\hat{\boldsymbol{z}}\\0&-2&3\\1&-2&0\end{pmatrix}=6\hat{\boldsymbol{x}}+3\hat{\boldsymbol{y}}+2\hat{\boldsymbol{z}}$$

The norm of the cross product is

$$|\boldsymbol{A} \times \boldsymbol{B}| = \sqrt{6^2 + 3^2 + 2^2} = 7$$

So the vector normal to the shaded plane is

$$\hat{\boldsymbol{n}} = \frac{1}{7}(6\hat{\boldsymbol{x}} + 3\hat{\boldsymbol{y}} + 2\hat{\boldsymbol{z}})$$

1.1.3 - Triple Products

Problem: 1.5

Prove the **BAC-CAB** rule by writing out both sides in component form.

Define

$$\begin{aligned}
\boldsymbol{A} &= A_x\hat{\boldsymbol{x}} + A_y\hat{\boldsymbol{y}} + A_z\hat{\boldsymbol{z}} \\
\boldsymbol{B} &= B_x\hat{\boldsymbol{x}} + B_y\hat{\boldsymbol{y}} + B_z\hat{\boldsymbol{z}} \\
\boldsymbol{C} &= C_x\hat{\boldsymbol{x}} + C_y\hat{\boldsymbol{y}} + C_z\hat{\boldsymbol{z}}
\end{aligned}$$

$\underline{\boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C})}$

$$\begin{aligned}
\boldsymbol{B} \times \boldsymbol{C} &= \det \begin{pmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{pmatrix} \\
&= (B_yC_z - B_zC_y)\,\hat{\boldsymbol{x}} + (B_zC_x - B_xC_z)\,\hat{\boldsymbol{y}} + (B_xC_y - B_yC_x)\,\hat{\boldsymbol{z}} \\
\boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C}) &= \det \begin{pmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ A_x & A_y & A_z \\ B_yC_z - B_zC_y & B_zC_x - B_xC_z & B_xC_y - B_yC_x \end{pmatrix} \\
&= \left[\underbrace{A_y(B_xC_y - B_yC_x)}_{x \text{ and } y} - \underbrace{A_z(B_zC_x - B_xC_z)}_{x \text{ and } z} \right] \hat{\boldsymbol{x}} \\
&\quad + \left[\underbrace{A_z(B_yC_z - B_zC_y)}_{y \text{ and } z} - \underbrace{A_x(B_xC_y - B_yC_x)}_{y \text{ and } x} \right] \hat{\boldsymbol{y}} \\
&\quad + \left[\underbrace{A_x(B_zC_x - B_xC_z)}_{z \text{ and } x} - \underbrace{A_y(B_yC_z - B_zC_y)}_{z \text{ and } y} \right] \hat{\boldsymbol{z}}
\end{aligned}$$

The variable pairings in each vector component are bracketed so as to expose the underlying structure: for the i-th unit vector, the corresponding i-components for all of $\boldsymbol{A}$, $\boldsymbol{B}$, and $\boldsymbol{C}$ are paired with the j- and k-components.

$\underline{\boldsymbol{B}(\boldsymbol{A}\cdot\boldsymbol{C})-\boldsymbol{C}(\boldsymbol{A}\cdot\boldsymbol{B})}$

$$
\begin{aligned}
\boldsymbol{B}(\boldsymbol{A}\cdot\boldsymbol{C}) &= (A_xC_x+A_yC_y+A_zC_z)(B_x\hat{\boldsymbol{x}}+B_y\hat{\boldsymbol{y}}+B_z\hat{\boldsymbol{z}}) \\
\boldsymbol{C}(\boldsymbol{A}\cdot\boldsymbol{B}) &= (A_xB_x+A_yB_y+A_zB_z)(C_x\hat{\boldsymbol{x}}+C_y\hat{\boldsymbol{y}}+C_z\hat{\boldsymbol{z}}) \\
\boldsymbol{B}(\boldsymbol{A}\cdot\boldsymbol{C})-\boldsymbol{C}(\boldsymbol{A}\cdot\boldsymbol{B}) &= \left[\underbrace{A_y(B_xC_y-B_yC_x)}_{x\text{ and }y}-\underbrace{A_z(B_zC_x-B_xC_z)}_{x\text{ and }z}\right]\hat{\boldsymbol{x}} \\
&\quad+\left[\underbrace{A_z(B_yC_z-B_zC_y)}_{y\text{ and }z}-\underbrace{A_x(B_xC_y-B_yC_x)}_{y\text{ and }x}\right]\hat{\boldsymbol{y}} \\
&\quad+\left[\underbrace{A_x(B_zC_x-B_xC_z)}_{z\text{ and }x}-\underbrace{A_y(B_yC_z-B_zC_y)}_{z\text{ and }y}\right]\hat{\boldsymbol{z}}
\end{aligned}
$$

Originally, we have eighteen terms before the subtraction. All terms with common components $(A_iB_iC_i)$ vanish (three in each of $\boldsymbol{B}(\boldsymbol{A}\cdot\boldsymbol{C})$ and $\boldsymbol{C}(\boldsymbol{A}\cdot\boldsymbol{B})$). We end up with twelve terms, as we should.

Problem: 1.6

Prove that

$$[\boldsymbol{A}\times(\boldsymbol{B}\times\boldsymbol{C})]+[\boldsymbol{B}\times(\boldsymbol{C}\times\boldsymbol{A})]+[\boldsymbol{C}\times(\boldsymbol{A}\times\boldsymbol{B})]=\boldsymbol{0}$$

Under what conditions does $\boldsymbol{A}\times(\boldsymbol{B}\times\boldsymbol{C})=(\boldsymbol{A}\times\boldsymbol{B})\times\boldsymbol{C}$?

Using the **BAC-CAB** identity and appealing to the commutativity of dot products, calculate

$$
\begin{aligned}
[\boldsymbol{A}\times(\boldsymbol{B}\times\boldsymbol{C})]+[\boldsymbol{B}\times(\boldsymbol{C}\times\boldsymbol{A})]+[\boldsymbol{C}\times(\boldsymbol{A}\times\boldsymbol{B})] &= \boldsymbol{B}(\boldsymbol{A}\cdot\boldsymbol{C})-\boldsymbol{C}(\boldsymbol{A}\cdot\boldsymbol{B}) \\
&\quad+\boldsymbol{C}(\boldsymbol{B}\cdot\boldsymbol{A})-\boldsymbol{A}(\boldsymbol{B}\cdot\boldsymbol{C}) \\
&\quad+\boldsymbol{A}(\boldsymbol{C}\cdot\boldsymbol{B})-\boldsymbol{B}(\boldsymbol{C}\cdot\boldsymbol{A}) \\
&= \boldsymbol{0}
\end{aligned}
$$

Equality holds when

$$\boldsymbol{B}\times(\boldsymbol{C}\times\boldsymbol{A})=\boldsymbol{0}$$

In this case, we can write $\boldsymbol{C}\times(\boldsymbol{A}\times\boldsymbol{B})=-(\boldsymbol{A}\times\boldsymbol{B})\times\boldsymbol{C}$, establishing the special case where associativity holds.

1.1.4 - Position, Displacement, and Separation Vectors

Problem: 1.7

Find the separation vector $\boldsymbol{\mathscr{r}}$ from the source point $(2,8,7)$ to the field point $(4,6,8)$. Determine its magnitude ($\mathscr{r}$), and construct the unit vector $\hat{\boldsymbol{\mathscr{r}}}$.

Compute
$$\boldsymbol{\mathscr{r}} = (4,6,8) - (2,8,7) = (2,-2,1)$$
which has magnitude
$$\mathscr{r} = \sqrt{2^2 + (-2)^2 + 1^2} = \sqrt{9} = 3$$
Thus the unit vector $\hat{\boldsymbol{\mathscr{r}}}$ is
$$\hat{\boldsymbol{\mathscr{r}}} = \frac{2}{3}\hat{\boldsymbol{x}} - \frac{2}{3}\hat{\boldsymbol{y}} + \frac{1}{3}\hat{\boldsymbol{z}}$$

1.1.5 - How Vectors Transform

Problem: 1.8

(a) Prove that the two-dimensional rotation matrix (Eq. 1.29) preserves dot products. (That is, show that $\overline{A}_y\overline{B}_y + \overline{A}_z\overline{B}_z = A_yB_y + A_zB_z$.)

(b) What constraints must the elements (R_{ij}) of the three-dimensional rotation matrix (Eq. 1.30) satisfy, in order to preserve the length of $\boldsymbol{A}$ (for all vectors $\boldsymbol{A}$)?

(a) Using the matrix rotation
$$\begin{pmatrix} \overline{A}_y \\ \overline{A}_z \end{pmatrix} = \begin{pmatrix} \cos(\phi) & \sin(\phi) \\ -\sin(\phi) & \cos(\phi) \end{pmatrix} \begin{pmatrix} A_y \\ A_z \end{pmatrix}$$
We have
$$\begin{aligned} \overline{A}_y &= A_y\cos(\phi) + A_z\sin(\phi) \\ \overline{A}_z &= -A_y\sin(\phi) + A_z\cos(\phi) \\ \overline{B}_y &= B_y\cos(\phi) + B_z\sin(\phi) \\ \overline{B}_z &= -B_y\sin(\phi) + B_z\cos(\phi) \end{aligned}$$
Multiply to derive
$$\begin{aligned} \overline{A}_y\overline{B}_y + \overline{A}_z\overline{B}_z &= A_yB_y\cos^2(\phi) + A_yB_z\cos(\phi)\sin(\phi) \\ &\quad + A_zB_y\cos(\phi)\sin(\phi) + A_zB_z\sin^2(\phi) \\ &\quad + A_yB_y\sin^2(\phi) - A_yB_z\cos(\phi)\sin(\phi) \\ &\quad - A_zB_y\cos(\phi)\sin(\phi) + A_zB_z\cos^2(\phi) \\ &= A_yB_y + A_zB_z \end{aligned}$$

(b) Using the matrix rotation

$$\begin{pmatrix}\overline{A}_x\\\overline{A}_y\\\overline{A}_z\end{pmatrix} = \begin{pmatrix}R_{xx} & R_{xy} & R_{xz}\\R_{yx} & R_{yy} & R_{yz}\\R_{zx} & R_{zy} & R_{zz}\end{pmatrix}\begin{pmatrix}A_x\\A_y\\A_z\end{pmatrix}$$

We have

$$\begin{aligned}\overline{A}_x &= A_xR_{xx} + A_yR_{xy} + A_zR_{xz}\\\overline{A}_y &= A_xR_{yx} + A_yR_{yy} + A_zR_{yz}\\\overline{A}_z &= A_xR_{zx} + A_yR_{zy} + A_zR_{zz}\end{aligned}$$

where a transformation is length-preserving if and only if $\sum_i \overline{A}_i^2 = \sum_i A_i^2$. We ought to be succinct and write in summation form

$$\sum_i \overline{A}_i^2 = \sum_i \left(\sum_j A_jR_{ij}\right)^2$$

It is rather tedious to evaluate these quantities, but writing $\overline{A}_x^2$ out identifies the pattern:

$$\begin{aligned}\overline{A}_x^2 = A_xR_{xx}(&A_xR_{xx} + A_yR_{xy} + A_zR_{xz})\\+ A_yR_{xy}(&A_xR_{xx} + A_yR_{xy} + A_zR_{xz})\\+ A_zR_{xz}(&A_xR_{xx} + A_yR_{xy} + A_zR_{xz})\end{aligned}$$

Equivalently, we write the summation

$$\sum_i \overline{A}_i^2 = \sum_j\sum_k\left(\sum_i R_{ij}R_{ik}\right)A_jA_k$$

Now, we need to force $\sum_j A_j^2$ on the right-hand side. One way to do so is to ensure that all products A_jA_k vanish. The first condition we impose is for each $j \neq k$,

$$\sum_i R_{ij}R_{ik} = 0$$

If you are familiar with the Kronecker delta notation, we can set the sum equal to δ_{jk}. Then the last question we are behooved to ask is what of the terms *with* $j = k$. It turns out the second condition is

$$\sum_i R_{ij}^2 = 1$$

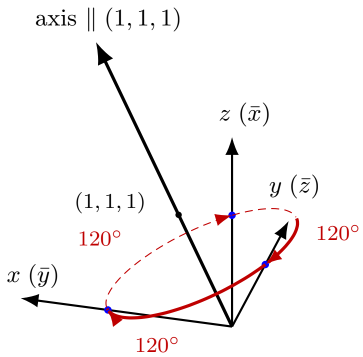
which leaves us with

$$\sum_i \overline{A}_i^2 = \sum_j A_j^2$$

Problem: 1.9

Find the transformation matrix R that describes a rotation by $120°$ about an axis from the origin through the point $(1, 1, 1)$. The rotation is clockwise as you look down the axis toward the origin.

An illustration is always helpful in this analysis.



rotation: $120°$ about the axis through $(1, 1, 1)$, clockwise as viewed looking down the axis toward O. Each unit vector slides $120°$ along the shared circle: $\hat{x} \to \hat{z}$, $\hat{z} \to \hat{y}$, $\hat{y} \to \hat{x}$, so the new axes are $\bar{x} \parallel z$, $\bar{y} \parallel x$, $\bar{z} \parallel y$.

The unique aspect of this situation is our axis of rotation is equidistant from each of the three axes. The axes trisect the circumference of the circular "orbit" about the axis of rotation at precisely $120°$ intervals – the same as the angle of rotation. Exploiting this symmetry, the key insight here is **rotation is equivalent to permutation**. The corresponding permutation matrix to the (231) rotation is

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

Problem: 1.10

(a) How do the components of a vector transform under a **translation** of coordinates ($\overline{x} = x$, $\overline{y} = y - a$, $\overline{z} = z$, Fig. 1.16a)?

(b) How do the components of a vector transform under an **inversion** of coordinates ($\overline{x} = -x$, $\overline{y} = -y$, $\overline{z} = -z$, Fig. 1.16b)?

(c) How do the components of a cross product (Eq. 1.13) transform under inversion? [The cross-product of two vectors is properly called a **pseudovector** because of this "anomalous" behavior.] Is the cross product of two pseudovectors a vector, or a pseudovector? Name two pseudovector quantities in classical mechanics.

(d) How does the scalar triple product of three vectors transform under inversions? (Such an object is called a **pseudoscalar**.)

(a) The physical position of the vector in space from the source to field points is **unchanged**. To evaluate this, consider the invariance of displacement in the y-component:

$$\overline{A}_y = \overline{y}_2 - \overline{y}_1 = (y_2 - a) - (y_1 - a) = y_2 - y_1 = A_y$$

Therefore the coordinates of the vector are unchanged.

(b) Once more, the physical position of the vector in space is invariant, but to reflect the inversion of the axes in preserving position, all components undergo a sign change:

$$\overline{A}_i = -A_i$$

(c) The anomaly lies in the fact that the components of a cross product **do not** change under inversion, even if the sign of the components to the inputs $\boldsymbol{A}$ and $\boldsymbol{B}$ vectors *did* flip. Put differently, $\boldsymbol{A}$ and $\boldsymbol{B}$ **do not** actually change their position in space, as it is only the axes that undergo inversion, and the signs of the components flip to reflect that. Concretely,

$$\begin{aligned}
\overline{\boldsymbol{A}} &= \overline{A}_x\hat{\boldsymbol{x}} + \overline{A}_y\hat{\boldsymbol{y}} + \overline{A}_z\hat{\boldsymbol{z}} \\
&= -A_x\hat{\boldsymbol{x}} - A_y\hat{\boldsymbol{y}} - A_z\hat{\boldsymbol{z}} \\
\overline{\boldsymbol{B}} &= \overline{B}_x\hat{\boldsymbol{x}} + \overline{B}_y\hat{\boldsymbol{y}} + \overline{B}_z\hat{\boldsymbol{z}} \\
&= -B_x\hat{\boldsymbol{x}} - B_y\hat{\boldsymbol{y}} - B_z\hat{\boldsymbol{z}} \\
\overline{\boldsymbol{A}} \times \overline{\boldsymbol{B}} &= \underbrace{[(-A_y)(-B_z) - (-A_z)(-B_y)]}_{\text{sign cancellation}}\hat{\boldsymbol{x}} + (A_zB_x - A_xB_z)\hat{\boldsymbol{y}} + (A_xB_y - A_yB_x)\hat{\boldsymbol{z}} \\
&= \boldsymbol{A} \times \boldsymbol{B}
\end{aligned}$$

Therein that derivation, we find a concise definition for a pseudovector: under inversion, we have

$$\overline{\boldsymbol{A}} = \boldsymbol{A}$$

Similarly, we consider two pseudovectors $\boldsymbol{C}$ and $\boldsymbol{D}$. Inverting to $\overline{\boldsymbol{C}}$ and $\overline{\boldsymbol{D}}$, by the same logic as above, the cross product of two pseudovectors is itself another pseudovector. Lastly, angular velocity and angular momentum are two pseudovector quantities in classical mechanics. Recall that the velocity of a point in a rotating body is given by

$$\boldsymbol{v} = \boldsymbol{\omega} \times \boldsymbol{r}$$

where $\boldsymbol{\omega}$ is the angular velocity and $\boldsymbol{r}$ the radial vector. Now invert all of the vectors to derive:

$$-\boldsymbol{v} = \overline{\boldsymbol{\omega}} \times (-\boldsymbol{r}) = -(\overline{\boldsymbol{\omega}} \times \boldsymbol{r}) \quad \Longrightarrow \boldsymbol{v} = \overline{\boldsymbol{\omega}} \times \boldsymbol{r}$$

We have demonstrated that $\overline{\boldsymbol{\omega}} = \boldsymbol{\omega}$ under inversion, ascertaining its pseudovector quality. For angular momentum $\boldsymbol{L}$, recall that the definition

$$\boldsymbol{L} = \boldsymbol{r} \times \boldsymbol{p}$$

lends itself to our previous derivation of $\overline{\boldsymbol{A}} \times \overline{\boldsymbol{B}} = \boldsymbol{A} \times \boldsymbol{B}$; simply replace the variables to see angular momentum is too a pseudovector.

(d) Under inversion, $\overline{\boldsymbol{A}} = -\boldsymbol{A}$ for any (real) vector $\boldsymbol{A}$. Derive

$$\begin{aligned} \overline{\boldsymbol{A}} \cdot \left(\overline{\boldsymbol{B}} \times \overline{\boldsymbol{C}}\right) &= \overline{\boldsymbol{A}} \cdot (\boldsymbol{B} \times \boldsymbol{C}) \\ &= -\boldsymbol{A} \cdot (\boldsymbol{B} \times \boldsymbol{C}) \end{aligned}$$

Unlike in the case of the pseudovector, the pseudoscalar **does** change sign.

1.2 - Differential Calculus

1.2.2 - Gradient

Problem: 1.11

Find the gradients of the following functions:

(a) $f(x, y, z) = x^2 + y^3 + z^4$

(b) $f(x, y, z) = x^2 y^3 z^4$

(c) $f(x, y, z) = e^x \sin(y)\ln(z)$

(a) $\nabla f = 2x\hat{\boldsymbol{x}} + 3y^2\hat{\boldsymbol{y}} + 4z^3\hat{\boldsymbol{z}}$

(b) $\nabla f = 2xy^3z^4\hat{\boldsymbol{x}} + 3x^2y^2z^4\hat{\boldsymbol{y}} + 4x^2y^3z^3\hat{\boldsymbol{z}}$

(c) $\nabla f = e^x\left[\sin(y)\ln(z)\,\hat{\boldsymbol{x}} + \cos(y)\ln(z)\,\hat{\boldsymbol{y}} + \sin(y)\,\frac{1}{z}\hat{\boldsymbol{z}}\right]$

Problem: 1.12

The height of a certain hill (in feet) is given by

$$h(x, y) = 10\big(2xy - 3x^2 - 4y^2 - 18x + 28y + 12\big)$$

where y is the distance (in miles) north, x the distance east of South Hadley.

(a) Where is the top of the hill located?

(b) How high is the hill?

(c) How steep is the slope (in feet per mile) at a point 1 mile north and one mile east of South Hadley? In what direction is the slope steepest, at that point?

(a) Calculate

$$\nabla h(x, y) = 10[(2y - 6x - 18)\,\hat{\boldsymbol{x}} + (2x - 8y + 28)\,\hat{\boldsymbol{y}}]$$

An extremum of the hill is located at (x, y) satisfying

$$\begin{pmatrix} -6 & 2 \\ 2 & -8 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 18 \\ -28 \end{pmatrix}$$

Which is solved by $(-2, 3)$. The Hessian and its determinant are given by

$$D = 10^2 \cdot \det\begin{pmatrix} -6 & 2 \\ 2 & -8 \end{pmatrix} = 4400$$

Since $D > 0$ and $f_{xx} = -6 < 0$, we can confirm that this is a local maximum.

(b)

$$h(-2,3) = 10\left[2(-2)(3) - 3(-2)^2 - 4(3)^2 - 18(-2) + 28(3) + 12\right] = 72 \text{ feet}$$

(c) Let $x = 1$ and $y = 1$. Then

$$\begin{aligned} |\nabla h(1,1)| &= \sqrt{10(2(1) - 6(1) - 18)^2 + 10(2(1) - 8(1) + 28)^2} \\ &= \sqrt{(-220)^2 + 220^2} = 220\sqrt{2} \\ &\approx 311 \text{ ft/mi} \end{aligned}$$

and the direction of the steepest slope points towards

$$\nabla h(1,1) = -220\hat{\boldsymbol{x}} + 220\hat{\boldsymbol{y}}$$

Problem: 1.13

Let $\boldsymbol{\mathscr{r}}$ be the separation vector from a fixed point (x', y', z') to the point (x, y, z) and let $\mathscr{r}$ be its length. Show that

(a) $\nabla(\mathscr{r}^2) = 2\boldsymbol{\mathscr{r}}$

(b) $\nabla(1/\mathscr{r}) = -\hat{\boldsymbol{\mathscr{r}}}/\mathscr{r}^2$

(c) What is the *general* formula for $\nabla(\mathscr{r}^n)$?

For each problem, make use of the following results:

$$\frac{\partial \mathscr{r}}{\partial i} = \frac{i - i'}{\mathscr{r}}, \quad i \in \{x, y, z\}$$

(a) The squared length of $\boldsymbol{\mathscr{r}}$ is

$$\mathscr{r}^2 = (x - x')^2 + (y - y')^2 + (z - z')^2$$

Then its gradient is

$$\begin{aligned} \nabla(\mathscr{r}^2) &= 2\mathscr{r}\left(\frac{\partial \mathscr{r}}{\partial x}\hat{\boldsymbol{x}} + \frac{\partial \mathscr{r}}{\partial y}\hat{\boldsymbol{y}} + \frac{\partial \mathscr{r}}{\partial z}\hat{\boldsymbol{z}}\right) \\ &= 2\mathscr{r}\left[\frac{(x - x')}{\mathscr{r}}\hat{\boldsymbol{x}} + \frac{(y - y')}{\mathscr{r}}\hat{\boldsymbol{y}} + \frac{(z - z')}{\mathscr{r}}\hat{\boldsymbol{z}}\right] \\ &= 2[(x - x')\hat{\boldsymbol{x}} + (y - y')\hat{\boldsymbol{y}} + (z - z')\hat{\boldsymbol{z}}] \\ &= 2\boldsymbol{\mathscr{r}} \end{aligned}$$

(b) The inverse length is

$$\frac{1}{\mathscr{r}} = \left[(x - x')^2 + (y - y')^2 + (z - z')^2\right]^{-1/2}$$

From this we calculate

$$\begin{aligned}\nabla\left(\frac{1}{\mathscr{r}}\right) &= -\frac{1}{\mathscr{r}^2}\left(\frac{\partial r}{\partial x}\hat{\boldsymbol{x}} + \frac{\partial r}{\partial y}\hat{\boldsymbol{y}} + \frac{\partial r}{\partial z}\hat{\boldsymbol{z}}\right)\\ &= -\frac{1}{\mathscr{r}^2}\left[\frac{(x-x')}{\mathscr{r}}\hat{\boldsymbol{x}} + \frac{(y-y')}{\mathscr{r}}\hat{\boldsymbol{y}} + \frac{(z-z')}{\mathscr{r}}\hat{\boldsymbol{z}}\right]\\ &= -\frac{\hat{\boldsymbol{\mathscr{r}}}}{\mathscr{r}^2}\end{aligned}$$

(c) For $n > 0$, derive each component

$$\frac{\partial}{\partial i}(\mathscr{r}^n) = n\mathscr{r}^{n-1}\frac{\partial \mathscr{r}}{\partial i}$$

which produces gradient

$$\nabla(\mathscr{r}^n) = n\mathscr{r}^{n-1}\hat{\boldsymbol{\mathscr{r}}}$$

Analogously, in the inverse case, each component is derived as

$$\frac{\partial}{\partial i}\left(\frac{1}{\mathscr{r}^n}\right) = -\frac{n}{\mathscr{r}^{n+1}}\frac{\partial r}{\partial i}$$

which corresponds to gradient

$$\nabla\left(\frac{1}{\mathscr{r}^n}\right) = -\frac{n}{\mathscr{r}^{n+1}}\hat{\boldsymbol{\mathscr{r}}}$$

We may actually combine these formulae for all $n \in \mathbb{R}$, and simply write

$$\nabla(\mathscr{r}^n) = n\mathscr{r}^{n-1}\hat{\boldsymbol{\mathscr{r}}}$$

Problem: 1.14

Suppose that f is a function of two variables (y and z) only. Show that the gradient $\nabla f = (\partial f/\partial y)\,\hat{\boldsymbol{y}} + (\partial f/\partial z)\,\hat{\boldsymbol{z}}$ transforms as a vector under rotations, Eq. 1.29. [*Hint*: $(\partial f/\partial \overline{y}) = (\partial f/\partial y)(\partial y/\partial \overline{y}) + (\partial f/\partial z)(\partial z/\partial \overline{y})$, and the analogous formula for $\partial f/\partial \overline{z}$. We know that $\overline{y} = y\cos(\phi) + z\sin(\phi)$ and $\overline{z} = -y\sin(\phi) + z\cos(\phi)$; "solve" these equations for y and z (as functions of $\overline{y}$ and $\overline{z}$, and compute the needed derivatives $\partial y/\partial \overline{y}$, $\partial z/\partial \overline{y}$, etc.]

The transformed coordinates $\overline{y}$ and $\overline{z}$ are defined as

$$\begin{pmatrix}\cos(\phi) & \sin(\phi)\\ -\sin(\phi) & \cos(\phi)\end{pmatrix}\begin{pmatrix}y\\ z\end{pmatrix} = \begin{pmatrix}\overline{y}\\ \overline{z}\end{pmatrix}$$

and y and z are solved for by

$$\begin{pmatrix}y\\ z\end{pmatrix} = \begin{pmatrix}\cos(\phi) & -\sin(\phi)\\ \sin(\phi) & \cos(\phi)\end{pmatrix}\begin{pmatrix}\overline{y}\\ \overline{z}\end{pmatrix}$$

$$y = \overline{y}\cos(\phi) - \overline{z}\sin(\phi)$$
$$z = \overline{y}\sin(\phi) + \overline{z}\cos(\phi)$$

The gradient in the transformed coordinate space takes on form

$$\nabla f = \frac{\partial f}{\partial \overline{y}}\hat{\boldsymbol{y}} + \frac{\partial f}{\partial \overline{z}}\hat{\boldsymbol{z}}$$

where by the chain rule

$$\begin{aligned}\frac{\partial f}{\partial \overline{y}} &= \frac{\partial f}{\partial y}\frac{\partial y}{\partial \overline{y}} + \frac{\partial f}{\partial z}\frac{\partial z}{\partial \overline{y}}\\ \frac{\partial f}{\partial \overline{z}} &= \frac{\partial f}{\partial y}\frac{\partial y}{\partial \overline{z}} + \frac{\partial f}{\partial z}\frac{\partial z}{\partial \overline{z}}\end{aligned}$$

From our equations for y and z, we calculate the partial derivatives with respect to the transformed coordinates as

$$\frac{\partial y}{\partial \overline{y}} = \cos(\phi)\,, \quad \frac{\partial y}{\partial \overline{z}} = -\sin(\phi)\,, \quad \frac{\partial z}{\partial \overline{y}} = \sin(\phi)\,, \quad \frac{\partial z}{\partial \overline{z}} = \cos(\phi)$$

Combining our results into the rotation matrix and corresponding matrix equation, we conclude that the gradient transforms as a vector under rotations; particularly, the rotation matrix is precisely the same as that in the transform from (y, z) to $(\overline{y}, \overline{z})$.

$$\begin{pmatrix}\partial f/\partial \overline{y}\\ \partial f/\partial \overline{z}\end{pmatrix} = \begin{pmatrix}\cos(\phi) & \sin(\phi)\\ -\sin(\phi) & \cos(\phi)\end{pmatrix}\begin{pmatrix}\partial f/\partial y\\ \partial f/\partial z\end{pmatrix}$$

1.2.4 - The Divergence

Problem: 1.15

Calculate the divergence of the following vector functions:

(a) $\boldsymbol{v}_a = x^2\hat{\boldsymbol{x}} + 3xz^2\hat{\boldsymbol{y}} - 2xz\hat{\boldsymbol{z}}$

(b) $\boldsymbol{v}_b = xy\hat{\boldsymbol{x}} + 2yz\hat{\boldsymbol{y}} + 3zx\hat{\boldsymbol{z}}$

(c) $\boldsymbol{v}_c = y^2\hat{\boldsymbol{x}} + \left(2xy + z^2\right)\hat{\boldsymbol{y}} + 2yz\hat{\boldsymbol{z}}$

(a) $\nabla \cdot \boldsymbol{v}_a = \frac{\partial}{\partial x}\left(x^2\right) + \frac{\partial}{\partial y}\left(3xz^2\right) - \frac{\partial}{\partial z}(2xz) = 2x - 2x = 0$

(b) $\nabla \cdot \boldsymbol{v}_b = \frac{\partial}{\partial x}(xy) + \frac{\partial}{\partial y}(2yz) + \frac{\partial}{\partial z}(3zx) = 3x + y + 2z$

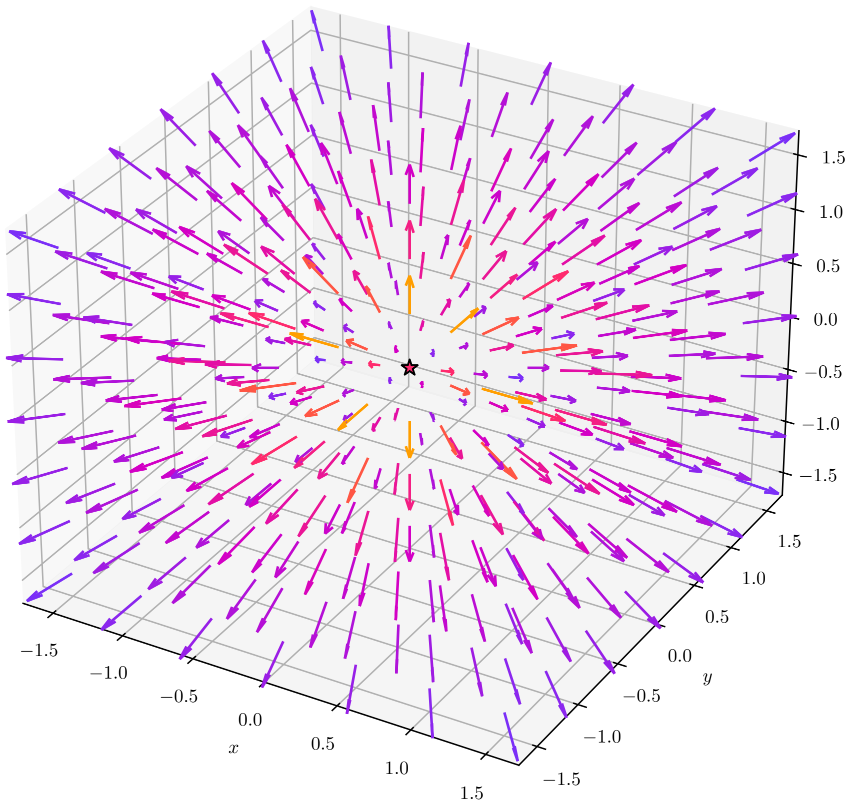
(c) $\boldsymbol{v}_c = \frac{\partial}{\partial x}\left(y^2\right) + \frac{\partial}{\partial y}\left(2xy + z^2\right) + \frac{\partial}{\partial z}(2yz) = 2x + 2y$

Problem: 1.16

Sketch the vector function
$$\boldsymbol{v} = \frac{\hat{\boldsymbol{r}}}{r^2}$$
and compute its divergence. The answer may surprise you...can you explain it?

We graph the vector function below:

$\mathbf{v} = \hat{\mathbf{r}}/r^2$ in 3D (color $= \log_{10} |\mathbf{v}|$)



Let
$$r = \sqrt{x^2 + y^2 + z^2}, \quad \hat{\boldsymbol{r}} = \frac{x\hat{\boldsymbol{x}} + y\hat{\boldsymbol{y}} + z\hat{\boldsymbol{z}}}{\sqrt{x^2 + y^2 + z^2}}$$
Then
$$\boldsymbol{v} = \frac{\hat{\boldsymbol{r}}}{r^2} = \frac{x\hat{\boldsymbol{x}} + y\hat{\boldsymbol{y}} + z\hat{\boldsymbol{z}}}{(x^2 + y^2 + z^2)^{3/2}}$$

The divergence is computed as

$$\begin{aligned}\nabla\cdot\boldsymbol{v} &= \frac{\partial}{\partial x}\left(\frac{x}{(x^2+y^2+z^2)^{3/2}}\right)+\frac{\partial}{\partial y}\left(\frac{y}{(x^2+y^2+z^2)^{3/2}}\right)\\ &\quad+\frac{\partial}{\partial z}\left(\frac{z}{(x^2+y^2+z^2)^{3/2}}\right)\\ &= \frac{\left(y^2+z^2-2x^2\right)+\left(x^2+z^2-2y^2\right)+\left(x^2+y^2-2z^2\right)}{(x^2+y^2+z^2)^{5/2}}\\ &= 0\end{aligned}$$

The counterintuitive solution, then, is the radial vector $\boldsymbol{v}$ **has vanishing divergence everywhere except the origin**, for this would produce $0/0$. The mechanics of the delta function later will enable us to explore this quandary deeper.

Problem: 1.17

In two dimensions, show that the divergence transforms as a scalar under rotations. [*Hint*: Use Eq. 1.29 to determine $\overline{v}_y$ and $\overline{v}_z$, and the method of Prob. 1.14 to calculate the derivatives. Your aim is to show that $\partial\overline{v}_y/\partial\overline{y}+\partial\overline{v}_z/\partial\overline{z}=\partial v_y/\partial y+\partial v_z/\partial z$.]

By our rotation matrix equation, the transformed quantities are

$$\begin{aligned}\overline{v}_y &= v_y\cos(\phi)+v_z\sin(\phi)\\ \overline{v}_z &= -v_y\sin(\phi)+v_z\cos(\phi)\end{aligned}$$

The inverse rotation takes us back to

$$\begin{aligned}v_y &= \overline{v}_y\cos(\phi)-\overline{v}_z\sin(\phi)\\ v_z &= \overline{v}_y\sin(\phi)+\overline{v}_z\cos(\phi)\end{aligned}$$

Using the chain rule, the derivation for $\partial\overline{v}_y/\partial\overline{y}$ and $\partial\overline{v}_z/\partial\overline{z}$ is

$$\begin{aligned}\frac{\partial\overline{v}_y}{\partial\overline{y}} &= \frac{\partial\overline{v}_y}{\partial v_y}\left(\frac{\partial v_y}{\partial y}\frac{\partial y}{\partial\overline{y}}+\frac{\partial v_y}{\partial z}\frac{\partial z}{\partial\overline{y}}\right)+\frac{\partial\overline{v}_y}{\partial v_z}\left(\frac{\partial v_z}{\partial y}\frac{\partial y}{\partial\overline{y}}+\frac{\partial v_z}{\partial z}\frac{\partial z}{\partial\overline{y}}\right)\\ \frac{\partial\overline{v}_z}{\partial\overline{z}} &= \frac{\partial\overline{v}_z}{\partial v_y}\left(\frac{\partial v_y}{\partial y}\frac{\partial y}{\partial\overline{z}}+\frac{\partial v_y}{\partial z}\frac{\partial z}{\partial\overline{z}}\right)+\frac{\partial\overline{v}_z}{\partial v_z}\left(\frac{\partial v_z}{\partial y}\frac{\partial y}{\partial\overline{z}}+\frac{\partial v_z}{\partial z}\frac{\partial z}{\partial\overline{z}}\right)\end{aligned}$$

From problem 1.14, we have each of $\partial y/\partial\overline{y}$, $\partial y/\partial\overline{z}$, $\partial z/\partial\overline{y}$, and $\partial z/\partial\overline{z}$. From the rotation equations above, we also have

$$\frac{\partial\overline{v}_y}{\partial v_y}=\cos(\phi)\,,\quad \frac{\partial\overline{v}_y}{\partial v_z}=\sin(\phi)\,,\quad \frac{\partial\overline{v}_z}{\partial v_y}=-\sin(\phi)\,,\quad \frac{\partial\overline{v}_z}{\partial v_z}=\cos(\phi)$$

Combining our ingredients and invoking the Pythagorean theorem, conclude

$$\begin{aligned}
\frac{\partial \bar{v}_y}{\partial \bar{y}} &= \frac{\partial v_y}{\partial y}\cos^2(\phi) + \frac{\partial v_y}{\partial z}\sin(\phi)\cos(\phi) \\
&\quad + \frac{\partial v_z}{\partial y}\sin(\phi)\cos(\phi) + \frac{\partial v_z}{\partial z}\sin^2(\phi) \\
\frac{\partial \bar{v}_z}{\partial \bar{z}} &= \frac{\partial v_z}{\partial z}\cos^2(\phi) - \frac{\partial v_y}{\partial z}\sin(\phi)\cos(\phi) \\
&\quad - \frac{\partial v_z}{\partial y}\sin(\phi)\cos(\phi) + \frac{\partial v_y}{\partial y}\sin^2(\phi) \\
\implies \frac{\partial \bar{v}_y}{\partial \bar{y}} + \frac{\partial \bar{v}_z}{\partial \bar{z}} &= \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}
\end{aligned}$$

1.2.5 - The Curl

Problem: 1.18

Calculate the curls of the vector functions in Prob. 1.15.

(a)

$$\begin{aligned}
\nabla \times \boldsymbol{v}_a &= \begin{vmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^2 & 3xz^2 & -2xz \end{vmatrix} \\
&= \left[\frac{\partial}{\partial y}(-2xz) - \frac{\partial}{\partial z}(3xz^2)\right]\hat{\boldsymbol{x}} + \left[\frac{\partial}{\partial z}(x^2) - \frac{\partial}{\partial x}(-2xz)\right]\hat{\boldsymbol{y}} \\
&\quad + \left[\frac{\partial}{\partial x}(3xz^2) - \frac{\partial}{\partial y}(x^2)\right]\hat{\boldsymbol{z}} \\
&= -6xz\hat{\boldsymbol{x}} + 2z\hat{\boldsymbol{y}} + 3z^2\hat{\boldsymbol{z}}
\end{aligned}$$

(b)

$$\begin{aligned}
\nabla \times \boldsymbol{v}_b &= \begin{vmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ xy & 2yz & 3zx \end{vmatrix} \\
&= \left[\frac{\partial}{\partial y}(3zx) - \frac{\partial}{\partial z}(2yz)\right]\hat{\boldsymbol{x}} + \left[\frac{\partial}{\partial z}(xy) - \frac{\partial}{\partial x}(3zx)\right]\hat{\boldsymbol{y}} \\
&\quad + \left[\frac{\partial}{\partial x}(2yz) - \frac{\partial}{\partial y}(xy)\right]\hat{\boldsymbol{z}} \\
&= -2y\hat{\boldsymbol{x}} - 3z\hat{\boldsymbol{y}} - x\hat{\boldsymbol{z}}
\end{aligned}$$

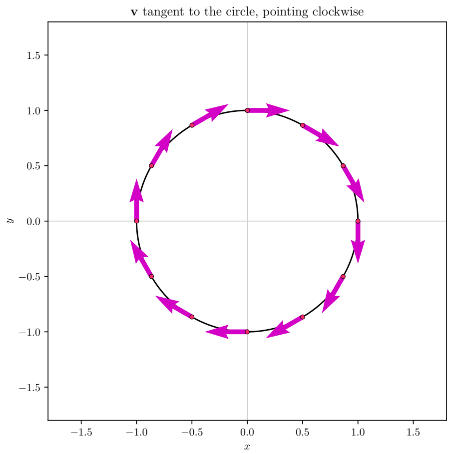
(c)

$$
\begin{aligned}
\nabla \times \boldsymbol{v}_c &= \begin{vmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ y^2 & 2xy + z^2 & 2yz \end{vmatrix} \\
&= \left[\frac{\partial}{\partial y}(2yz) - \frac{\partial}{\partial z}(2xy + z^2)\right] \hat{\boldsymbol{x}} + \left[\frac{\partial}{\partial z}(y^2) - \frac{\partial}{\partial x}(2yz)\right] \hat{\boldsymbol{y}} \\
&\quad + \left[\frac{\partial}{\partial x}(2xy + z^2) - \frac{\partial}{\partial y}(y^2)\right] \\
&= \mathbf{0}
\end{aligned}
$$

Problem: 1.19

Draw a circle in the xy plane. At a few representative points draw the vector $\boldsymbol{v}$ tangent to the circle, pointing in the clockwise direction. By comparing adjacent vectors, determine the *sign* of $\partial v_x/\partial y$ and $\partial v_y/\partial x$. According to Eq. 1.41, then, what is the direction of $\nabla \times \boldsymbol{v}$? Explain how this example illustrates the geometrical interpretation of the curl.

We illustrate tangent vectors pointing clockwise on the unit circle:



Geometrically, observe how as x increases, the v_y component decreases, and as y increases, v_x increases. Thus $\partial v_x/\partial y > 0$ and $\partial v_y/\partial x < 0$.

Analytically, the vector field describing these tangent clockwise vectors is

$$\boldsymbol{v} = v_x\hat{\boldsymbol{x}} + v_y\hat{\boldsymbol{y}} = y\hat{\boldsymbol{x}} - x\hat{\boldsymbol{y}}$$

Meaning that $\partial v_x/\partial y > 0$ and $\partial v_y/\partial x < 0$, and thus $\partial v_y/\partial x - \partial v_x/\partial y < 0$. This suggests that the curl points in the **negative** z-direction.

$$\nabla \times \boldsymbol{v} = \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y}\right)\hat{\boldsymbol{z}} = -2\hat{\boldsymbol{z}}$$

By the right-handedness of the coordinate system, this is geometrically intuitive. From our computation of $\nabla \times \boldsymbol{v}$, we see that the curl is in the negative z-direction *everywhere*, indicating that if we took any point and applied the right-hand rule, our thumb would point downwards everywhere.

Problem: 1.20

Construct a vector function that has zero divergence and zero curl everywhere. (A *constant* will do the job, of course, but make it something a little more interesting than that!)

Let $\boldsymbol{v} = v_x\hat{\boldsymbol{x}} + v_y\hat{\boldsymbol{y}} + v_z\hat{\boldsymbol{z}}$. We must find a vector $\boldsymbol{v}$ whose components satisfy these relations:

$$\begin{aligned} \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} &= 0 \\ \frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} &= 0 \\ \frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} &= 0 \\ \frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} &= 0 \end{aligned}$$

One insight is to realize that if each of the three components are purely in terms of x, y, and z, respectively, then immediately the curl goes to zero. Suppose $v_x = f(x)$, $v_y = g(y)$, and $v_z = h(z)$. Then we need only find functions that satisfy

$$\frac{df}{dx} + \frac{dg}{dy} + \frac{dh}{dz} = 0$$

An easy example is: $f(x) = -x$, $g(y) = -y$, and $h(z) = 2z$.

1.2.6 - Product Rules

Problem: 1.21

Prove product rules (i), (iv), and (v).

(i) $\nabla(fg) = f\nabla g + g\nabla f$.

Proof.

$$\begin{aligned} \nabla(fg) &= \frac{\partial}{\partial x}(fg)\,\hat{\boldsymbol{x}} + \frac{\partial}{\partial y}(fg)\,\hat{\boldsymbol{y}} + \frac{\partial}{\partial z}(fg)\,\hat{\boldsymbol{z}} \\ &= \left(f\frac{\partial g}{\partial x} + g\frac{\partial f}{\partial x}\right)\hat{\boldsymbol{x}} + \left(f\frac{\partial g}{\partial y} + g\frac{\partial f}{\partial y}\right)\hat{\boldsymbol{y}} + \left(f\frac{\partial g}{\partial z} + g\frac{\partial f}{\partial z}\right)\hat{\boldsymbol{z}} \\ &= f\nabla g + g\nabla f \end{aligned}$$

□

(iv) $\nabla\cdot(\boldsymbol{A}\times\boldsymbol{B}) = \boldsymbol{B}\cdot(\nabla\times\boldsymbol{A}) - \boldsymbol{A}\cdot(\nabla\times\boldsymbol{B})$.

Proof. First compute $\boldsymbol{A}\times\boldsymbol{B}$:

$$\begin{aligned}\boldsymbol{A}\times\boldsymbol{B} &= \begin{vmatrix}\hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}}\\ A_x & A_y & A_z\\ B_x & B_y & B_z\end{vmatrix}\\ &= (A_yB_z - A_zB_y)\,\hat{\boldsymbol{x}} + (A_zB_x - A_xB_z)\,\hat{\boldsymbol{y}} + (A_xB_y - A_yB_x)\,\hat{\boldsymbol{z}}\end{aligned}$$

Then

$$\begin{aligned}\nabla\cdot(\boldsymbol{A}\times\boldsymbol{B}) &= \frac{\partial}{\partial x}(A_yB_z - A_zB_y) + \frac{\partial}{\partial y}(A_zB_x - A_xB_z) + \frac{\partial}{\partial z}(A_xB_y - A_yB_x)\\ &= A_y\frac{\partial B_z}{\partial x} + B_z\frac{\partial A_y}{\partial x} - A_z\frac{\partial B_y}{\partial x} - B_y\frac{\partial A_z}{\partial x}\\ &\quad + A_z\frac{\partial B_x}{\partial y} + B_x\frac{\partial A_z}{\partial y} - A_x\frac{\partial B_z}{\partial y} - B_z\frac{\partial A_x}{\partial y}\\ &\quad\quad + A_x\frac{\partial B_y}{\partial z} + B_y\frac{\partial A_x}{\partial z} - A_y\frac{\partial B_x}{\partial z} - B_x\frac{\partial A_y}{\partial z}\\ &= \underbrace{B_x\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right) + B_y\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right) + B_z\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)}_{\boldsymbol{B}\cdot(\nabla\times\boldsymbol{A})}\\ &\quad + \underbrace{A_x\left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z}\right) + A_y\left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x}\right) + A_z\left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y}\right)}_{\boldsymbol{A}\cdot(\nabla\times\boldsymbol{B})}\end{aligned}$$

□

(v) $\nabla\times(f\boldsymbol{A}) = f(\nabla\times\boldsymbol{A}) - \boldsymbol{A}\times(\nabla f)$.

Proof.

$$\begin{aligned}\nabla\times(f\boldsymbol{A}) &= \begin{vmatrix}\hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}}\\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z}\\ fA_x & fA_y & fA_z\end{vmatrix}\\ &= \left[\frac{\partial}{\partial y}(fA_z) - \frac{\partial}{\partial z}(fA_y)\right]\hat{\boldsymbol{x}} + \left[\frac{\partial}{\partial z}(fA_x) - \frac{\partial}{\partial x}(fA_z)\right]\hat{\boldsymbol{y}}\\ &\quad + \left[\frac{\partial}{\partial x}(fA_y) - \frac{\partial}{\partial y}(fA_x)\right]\hat{\boldsymbol{z}}\\ &= \left[f\frac{\partial A_z}{\partial y} + A_z\frac{\partial f}{\partial y} - f\frac{\partial A_y}{\partial z} - A_y\frac{\partial f}{\partial z}\right]\hat{\boldsymbol{x}}\\ &\quad + \left[f\frac{\partial A_x}{\partial z} + A_x\frac{\partial f}{\partial z} - f\frac{\partial A_z}{\partial x} - A_z\frac{\partial f}{\partial x}\right]\hat{\boldsymbol{y}}\\ &\quad\quad + \left[f\frac{\partial A_y}{\partial x} + A_y\frac{\partial f}{\partial x} - f\frac{\partial A_x}{\partial y} - A_x\frac{\partial f}{\partial y}\right]\hat{\boldsymbol{z}}\\ &= \underbrace{f\left[\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right)\hat{\boldsymbol{x}} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)\hat{\boldsymbol{y}} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_y}{\partial z}\right)\hat{\boldsymbol{z}}\right]}_{f(\nabla\times\boldsymbol{A})}\\ &\quad - \underbrace{\left[\left(A_y\frac{\partial f}{\partial z} - A_z\frac{\partial f}{\partial y}\right)\hat{\boldsymbol{x}} + \left(A_x\frac{\partial f}{\partial z} - A_z\frac{\partial f}{\partial x}\right)\hat{\boldsymbol{y}} + \left(A_x\frac{\partial f}{\partial y} - A_y\frac{\partial f}{\partial x}\hat{\boldsymbol{z}}\right)\right]}_{\boldsymbol{A}\times(\nabla f)}\end{aligned}$$

$\square$

Problem: 1.22

(a) If $\boldsymbol{A}$ and $\boldsymbol{B}$ are two vector functions, what does the expression $(\boldsymbol{A} \cdot \nabla) \boldsymbol{B}$ mean? (That is, what are its x, y, and z components, in terms of the Cartesian components of $\boldsymbol{A}$, $\boldsymbol{B}$, and ∇?)

(b) Compute $(\hat{\boldsymbol{r}} \cdot \nabla) \hat{\boldsymbol{r}}$, where $\hat{\boldsymbol{r}}$ is the unit vector defined in Eq. 1.21.

(c) For the functions in Prob. 1.15, evaluate $(\boldsymbol{v}_a \cdot \nabla) \boldsymbol{v}_b$.

(a) Define $\boldsymbol{A} = A_x\hat{\boldsymbol{x}} + A_y\hat{\boldsymbol{y}} + A_z\hat{\boldsymbol{z}}$ and $\boldsymbol{B} = B_x\hat{\boldsymbol{x}} + B_y\hat{\boldsymbol{y}} + B_z\hat{\boldsymbol{z}}$. Then

$$
\begin{aligned}
(\boldsymbol{A} \cdot \nabla) \boldsymbol{B} &= A_x \frac{\partial \boldsymbol{B}}{\partial x} + A_y \frac{\partial \boldsymbol{B}}{\partial y} + A_z \frac{\partial \boldsymbol{B}}{\partial z} \\
&= \sum_{i \in \{x,y,z\}} \left(\sum_{j \in \{x,y,z\}} A_j \frac{\partial B_i}{\partial j} \right) \hat{\boldsymbol{i}}
\end{aligned}
$$

We call this expression the **directional derivative**.

(b) For $i, j \in \{x, y, z\}$, derive

$$
\frac{\partial r_i}{\partial j} = \begin{cases} \dfrac{1}{r} - \dfrac{j^2}{r^3} & i = j \\ -\dfrac{ij}{r^3} & i \neq j \end{cases}
$$

Then we can compute

$$
(\hat{\boldsymbol{r}} \cdot \nabla) \hat{\boldsymbol{r}} = \frac{1}{r} \sum_{i \in \{x,y,z\}} \left(\sum_{j \in \{x,y,z\}} j \frac{\partial r_i}{\partial j} \right) \hat{\boldsymbol{i}}
$$

Take $i = x$ to see how this computation works. We have

$$
\begin{aligned}
[(\hat{\boldsymbol{r}} \cdot \nabla) \hat{\boldsymbol{r}}]_x &= \frac{1}{r} \left(x \frac{\partial r_x}{\partial x} + y \frac{\partial r_x}{\partial y} + z \frac{\partial r_x}{\partial z} \right) \\
&= \frac{1}{r} \left[x \left(\frac{1}{r} - \frac{x^2}{r^3} \right) - \frac{xy^2}{r^3} - \frac{xz^2}{r^3} \right] \\
&= \frac{1}{r} \left[x \left(\frac{1}{r} - \left(\frac{x^2 + y^2 + z^2}{r^3} \right) \right) \right] \\
&= \frac{1}{r} \left[x \left(\frac{1}{r} - \frac{1}{r} \right) \right] \\
&= 0
\end{aligned}
$$

An extraordinary reduction. By symmetry, it follows that the components for y and z follow analogously. Then we can conclude

$$
(\hat{\boldsymbol{r}} \cdot \nabla) \hat{\boldsymbol{r}} = \boldsymbol{0}
$$

(c)

$$
\begin{aligned}
(\boldsymbol{v}_a \cdot \nabla)\, \boldsymbol{v}_b &= \sum_{i \in \{x,y,z\}} \left(\sum_{j \in \{x,y,z\}} v_{a,j} \frac{\partial v_{b,i}}{\partial j} \right) \hat{\boldsymbol{\imath}} \\
&= \left(x^2 y + 3x^2 z^2\right) \hat{\boldsymbol{x}} + \left(6xz^3 - 4xyz\right) \hat{\boldsymbol{y}} - 3x^2 z \hat{\boldsymbol{z}}
\end{aligned}
$$

Problem: 1.23

(For masochists only.) Prove product rules (ii) and (vi). Refer to Prob. 1.22 for the definition of $(\boldsymbol{A} \cdot \nabla)\, \boldsymbol{B}$.

(ii) $\nabla(\boldsymbol{A} \cdot \boldsymbol{B}) = \boldsymbol{A} \times (\nabla \times \boldsymbol{B}) + \boldsymbol{B} \times (\nabla \times \boldsymbol{A}) + (\boldsymbol{A} \cdot \nabla)\, \boldsymbol{B} + (\boldsymbol{B} \cdot \nabla)\, \boldsymbol{A}$.

Proof.

$$
\begin{aligned}
\nabla(\boldsymbol{A} \cdot \boldsymbol{B}) &= \nabla(A_x B_x + A_y B_y + A_z B_z) \\
&= \sum_{i \in \{x,y,z\}} \left(\sum_{j \in \{x,y,z\}} A_j \frac{\partial B_j}{\partial i} + B_j \frac{\partial A_j}{\partial i} \right) \hat{\boldsymbol{\imath}} \\
\boldsymbol{A} \times (\nabla \times \boldsymbol{B}) &= \begin{vmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ A_x & A_y & A_z \\ \frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} & \frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} & \frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \end{vmatrix} \\
&= \left[A_y \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \right) - A_z \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} \right) \right] \hat{\boldsymbol{x}} \\
&\quad + \left[A_z \left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} \right) - A_x \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \right) \right] \hat{\boldsymbol{y}} \\
&\quad + \left[A_x \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} \right) - A_y \left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} \right) \right] \hat{\boldsymbol{z}} \\
\boldsymbol{B} \times (\nabla \times \boldsymbol{A}) &= \begin{vmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ B_x & B_y & B_z \\ \frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} & \frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} & \frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \end{vmatrix} \\
&= \left[B_y \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) - B_z \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \right] \hat{\boldsymbol{x}} \\
&\quad + \left[B_z \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) - B_x \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \right] \hat{\boldsymbol{y}} \\
&\quad + \left[B_x \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) - B_y \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \right] \hat{\boldsymbol{z}} \\
(\boldsymbol{A} \cdot \nabla)\, \boldsymbol{B} &= \sum_{i \in \{x,y,z\}} \left(\sum_{j \in \{x,y,z\}} A_j \frac{\partial B_i}{\partial j} \right) \hat{\boldsymbol{\imath}} \\
(\boldsymbol{B} \cdot \nabla)\, \boldsymbol{A} &= \sum_{i \in \{x,y,z\}} \left(\sum_{j \in \{x,y,z\}} B_j \frac{\partial A_i}{\partial j} \right) \hat{\boldsymbol{\imath}}
\end{aligned}
$$

Adding $\boldsymbol{A} \times (\nabla \times \boldsymbol{B})$ and $\boldsymbol{B} \times (\nabla \times \boldsymbol{A})$, derive

$$\sum_{i\in\{x,y,z\}} \left(\sum_{j\in\{x,y,z\}} (1-\delta_{ij}) \left(A_j \frac{\partial B_j}{\partial i} + B_j \frac{\partial A_j}{\partial i} \right) \right) \hat{\boldsymbol{i}}$$
$$- \sum_{i\in\{x,y,z\}} \left(\sum_{j\in\{x,y,z\}} (1-\delta_{ij}) \left(A_j \frac{\partial B_i}{\partial j} + B_j \frac{\partial A_i}{\partial j} \right) \right) \hat{\boldsymbol{i}}$$

Now, looking at the directional derivatives, we may write their sum $(\boldsymbol{A}\cdot\nabla)\boldsymbol{B} + (\boldsymbol{B}\cdot\nabla)\boldsymbol{A}$ as

$$\sum_{i\in\{x,y,z\}} \left(A_i \frac{\partial B_i}{\partial i} + B_i \frac{\partial A_i}{\partial i} \right) \hat{\boldsymbol{i}} + \sum_{i\in\{x,y,z\}} \left(\sum_{j\in\{x,y,z\}} (1-\delta_{ij}) \left(A_j \frac{\partial B_i}{\partial j} + B_j \frac{\partial A_i}{\partial j} \right) \right) \hat{\boldsymbol{i}}$$

The blue summations cancel each other out, while the red terms sum to the desired result. Observe that the heart of the proof arose in decomposing the derivative structures: the first corresponds to the gradient and the second to the directional derivative.

$$A_j \frac{\partial B_j}{\partial i} \text{ (gradient)} \qquad A_j \frac{\partial B_i}{\partial j} \text{ (directional derivative)}$$

□

(vi) $\nabla \times (\boldsymbol{A} \times \boldsymbol{B}) = (\boldsymbol{B}\cdot\nabla)\boldsymbol{A} - (\boldsymbol{A}\cdot\nabla)\boldsymbol{B} + \boldsymbol{A}(\nabla\cdot\boldsymbol{B}) - \boldsymbol{B}(\nabla\cdot\boldsymbol{A})$.

Proof.

$$\nabla \times (\boldsymbol{A} \times \boldsymbol{B}) = \begin{vmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_yB_z - A_zB_y & A_zB_x - A_xB_z & A_xB_y - A_yB_x \end{vmatrix}$$
$$= \sum_{i\in\{x,y,z\}} \left\{ \sum_{j\in\{x,y,z\}} \left[\left(A_i \frac{\partial B_j}{\partial j} + B_j \frac{\partial A_i}{\partial j} \right) - \left(A_j \frac{\partial B_i}{\partial j} + B_i \frac{\partial A_j}{\partial j} \right) \right] \right\} \hat{\boldsymbol{i}}$$

$$(\boldsymbol{B}\cdot\nabla)\boldsymbol{A} = \sum_{i\in\{x,y,z\}} \left(\sum_{j\in\{x,y,z\}} B_j \frac{\partial A_i}{\partial j} \right) \hat{\boldsymbol{i}}$$

$$(\boldsymbol{A}\cdot\nabla)\boldsymbol{B} = \sum_{i\in\{x,y,z\}} \left(\sum_{j\in\{x,y,z\}} A_j \frac{\partial B_i}{\partial j} \right) \hat{\boldsymbol{i}}$$

$$\boldsymbol{A}(\nabla\cdot\boldsymbol{B}) = \sum_{i\in\{x,y,z\}} \left(\sum_{j\in\{x,y,z\}} A_i \frac{\partial B_j}{\partial j} \right) \hat{\boldsymbol{i}}$$

$$\boldsymbol{B}(\nabla\cdot\boldsymbol{A}) = \sum_{i\in\{x,y,z\}} \left(\sum_{j\in\{x,y,z\}} B_i \frac{\partial A_j}{\partial j} \right) \hat{\boldsymbol{i}}$$

Where once again, we combine divergence and directional derivative components to produce the desired result. □

Problem: 1.24

Derive the three quotient rules.

Switch either $g \to g^{-1}$ or $f \to f^{-1}$ to find

$$\begin{aligned}
\nabla\left(\frac{f}{g}\right) &= \nabla\left(fg^{-1}\right) \\
&= f\nabla g^{-1} + g^{-1}\nabla f \\
&= \frac{g\nabla f - f\nabla g}{g^2} \\
\nabla\cdot\left(\frac{1}{f}\boldsymbol{A}\right) &= \nabla\cdot\left(f^{-1}\boldsymbol{A}\right) \\
&= f^{-1}(\nabla\cdot\boldsymbol{A}) + \boldsymbol{A}\cdot\left(\nabla f^{-1}\right) \\
&= \frac{f(\nabla\cdot\boldsymbol{A}) - \boldsymbol{A}\cdot\nabla f}{f^2} \\
\nabla\times\left(\frac{1}{f}\boldsymbol{A}\right) &= \nabla\times\left(f^{-1}\boldsymbol{A}\right) \\
&= f^{-1}(\nabla\times\boldsymbol{A}) - \boldsymbol{A}\times\left(\nabla f^{-1}\right) \\
&= \frac{f(\nabla\times\boldsymbol{A}) + \boldsymbol{A}\times\nabla f}{f^2}
\end{aligned}$$

Problem: 1.25

(a) Check product rule (iv) (by calculating each term separately) for the functions

$$\boldsymbol{A} = x\hat{\boldsymbol{x}} + 2y\hat{\boldsymbol{y}} + 3z\hat{\boldsymbol{z}} \qquad \boldsymbol{B} = 3y\hat{\boldsymbol{x}} - 2x\hat{\boldsymbol{y}}$$

(b) Do the same for product rule (ii).

(c) Do the asme for rule (vi).

Use the fact that $\nabla\times A = \mathbf{0}$ from our insight in problem 1.20 that a zero curl arises when each component function is strictly in the variable of its corresponding basis coordinate.

(a)

$$\begin{aligned}
\boldsymbol{A}\times\boldsymbol{B} &= \begin{vmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ x & 2y & 3z \\ 3y & -2x & 0 \end{vmatrix} \\
&= 6xz\hat{\boldsymbol{x}} + 9yz\hat{\boldsymbol{y}} - \left(2x^2+6y^2\right)\hat{\boldsymbol{z}} \\
\boxed{\nabla\cdot(\boldsymbol{A}\times\boldsymbol{B})} &= \frac{\partial}{\partial x}(6xz) + \frac{\partial}{\partial y}(9yz) - \frac{\partial}{\partial z}\left(2x^2+6y^2\right) \\
&= \boxed{15z} \\
\boxed{\boldsymbol{B}\cdot(\nabla\times\boldsymbol{A})} &= \boxed{0} \\
\nabla\times\boldsymbol{B} &= \begin{vmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3y & -2x & 0 \end{vmatrix} \\
&= -5\hat{\boldsymbol{z}} \\
\boxed{\boldsymbol{A}\cdot(\nabla\times\boldsymbol{B})} &= \boxed{-15z} \\
\boxed{\boldsymbol{B}\cdot(\nabla\times\boldsymbol{A}) - \boldsymbol{A}\cdot(\nabla\times\boldsymbol{B})} &= \boxed{15z}
\end{aligned}$$

(b)

$$\begin{aligned}
\boxed{\nabla(\boldsymbol{A}\cdot\boldsymbol{B})} &= \frac{\partial}{\partial x}(-xy)\,\hat{\boldsymbol{x}} + \frac{\partial}{\partial y}(-xy)\,\hat{\boldsymbol{y}} \\
&= \boxed{-y\hat{\boldsymbol{x}} - x\hat{\boldsymbol{y}}} \\
\nabla\times\boldsymbol{B} &= -5\hat{\boldsymbol{z}} \\
\boxed{\boldsymbol{A}\times(\nabla\times\boldsymbol{B})} &= \begin{vmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ x & 2y & 3z \\ 0 & 0 & -5 \end{vmatrix} \\
&= \boxed{-10y\hat{\boldsymbol{x}} + 5x\hat{\boldsymbol{y}}} \\
\boxed{\boldsymbol{B}\times(\nabla\times\boldsymbol{A})} &= \boxed{\boldsymbol{0}} \\
\boxed{(\boldsymbol{A}\cdot\nabla)\,\boldsymbol{B}} &= x\frac{\partial\boldsymbol{B}}{\partial x} + 2y\frac{\partial\boldsymbol{B}}{\partial y} \\
&= \boxed{6y\hat{\boldsymbol{x}} - 2x\hat{\boldsymbol{y}}} \\
\boxed{(\boldsymbol{B}\cdot\nabla)\,\boldsymbol{A}} &= 3y\frac{\partial\boldsymbol{A}}{\partial x} - 2x\frac{\partial\boldsymbol{A}}{\partial y} \\
&= \boxed{3y\hat{\boldsymbol{x}} - 4x\hat{\boldsymbol{y}}}
\end{aligned}$$

Summing our terms together, derive

$$\boldsymbol{A}\times(\nabla\times\boldsymbol{B}) + \boldsymbol{B}\times(\nabla\times\boldsymbol{A}) + (\boldsymbol{A}\cdot\nabla)\,\boldsymbol{B} + (\boldsymbol{B}\cdot\nabla)\,\boldsymbol{A} = \boxed{-y\hat{\boldsymbol{x}} - x\hat{\boldsymbol{y}}}$$

(c)

$$\boxed{\nabla \times (\boldsymbol{A} \times \boldsymbol{B})} = \begin{vmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 6xz & 9yz & -(2x^2 + 6y^2) \end{vmatrix}$$

$$= \left[\frac{\partial}{\partial y}\left[-(2x^2+6y^2)\right] - \frac{\partial}{\partial z}(9yz)\right]\hat{\boldsymbol{x}}$$

$$+ \left[\frac{\partial}{\partial z}(6xz) + \frac{\partial}{\partial x}(2x^2+6y^2)\right]\hat{\boldsymbol{y}}$$

$$+ \left[\frac{\partial}{\partial x}(9yz) - \frac{\partial}{\partial y}(6xz)\right]\hat{\boldsymbol{z}}$$

$$= \boxed{-21y\hat{\boldsymbol{x}} + 10x\hat{\boldsymbol{y}}}$$

$$\boxed{(\boldsymbol{B} \cdot \nabla)\,\boldsymbol{A}} = \boxed{3y\hat{\boldsymbol{x}} - 4x\hat{\boldsymbol{y}}}$$

$$\boxed{(\boldsymbol{A} \cdot \nabla)\,\boldsymbol{B}} = \boxed{6y\hat{\boldsymbol{x}} - 2x\hat{\boldsymbol{y}}}$$

$$\boxed{\boldsymbol{A}(\nabla \cdot \boldsymbol{B})} = \boxed{\boldsymbol{0}}$$

$$\boxed{\boldsymbol{B}(\nabla \cdot \boldsymbol{A})} = 6\boldsymbol{B}$$

$$= \boxed{18y\hat{\boldsymbol{x}} - 12x\hat{\boldsymbol{y}}}$$

Combining the last four terms as rule (vi) stipulates, we have

$$(\boldsymbol{B} \cdot \nabla)\,\boldsymbol{A} - (\boldsymbol{A} \cdot \nabla)\,\boldsymbol{B} + \boldsymbol{A}(\nabla \cdot \boldsymbol{B}) - \boldsymbol{B}(\nabla \cdot \boldsymbol{A}) = -21y\hat{\boldsymbol{x}} + 10x\hat{\boldsymbol{y}}$$

1.2.7 - Second Derivatives

Problem: 1.26

Calculate the Laplacian of the following functions:

(a) $T_a = x^2 + 2xy + 3z + 4$

(b) $T_b = \sin(x)\sin(y)\sin(z)$

(c) $T_c = e^{-5x}\sin(4y)\cos(3z)$

(d) $\boldsymbol{v} = x^2\hat{\boldsymbol{x}} + 3xz^2\hat{\boldsymbol{y}} - 2xz\hat{\boldsymbol{z}}$

(a)

$$\nabla^2 T_a = \frac{\partial^2 T_a}{\partial x^2} + \frac{\partial^2 T_a}{\partial y^2} + \frac{\partial^2 T_a}{\partial z^2} = 2$$

(b)

$$\nabla^2 T_b = \frac{\partial^2 T_b}{\partial x^2} + \frac{\partial^2 T_b}{\partial y^2} + \frac{\partial^2 T_b}{\partial z^2} = -3\sin(x)\sin(y)\sin(z)$$

(c)

$$\nabla^2 T_c = \frac{\partial^2 T_c}{\partial x^2} + \frac{\partial^2 T_c}{\partial y^2} + \frac{\partial^2 T_c}{\partial z^2} = (25 - 16 - 3)\,e^{-5x}\sin(4y)\cos(3z) = 0$$

(d)
$$\nabla^2 \boldsymbol{v} = \nabla^2\left(x^2\right)\hat{\boldsymbol{x}} + \nabla^2\left(3xz^2\right)\hat{\boldsymbol{y}} + \nabla^2(-2xz)\hat{\boldsymbol{z}} = 2\hat{\boldsymbol{x}} + 6x\hat{\boldsymbol{y}}$$

Problem: 1.27

Prove that the divergence of a curl is always zero. *Check* it for function $\boldsymbol{v}_a$ in Prob. 1.15.

$\nabla \cdot (\nabla \times \boldsymbol{v}) = 0.$

Proof.

$$\begin{aligned}
\nabla \times \boldsymbol{v} &= \begin{vmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_x & v_y & v_z \end{vmatrix} \\
&= \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z}\right)\hat{\boldsymbol{x}} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x}\right)\hat{\boldsymbol{y}} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y}\right)\hat{\boldsymbol{z}} \\
\nabla \cdot (\nabla \times \boldsymbol{v}) &= \frac{\partial}{\partial x}\left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z}\right) + \frac{\partial}{\partial y}\left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x}\right) + \frac{\partial}{\partial z}\left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y}\right)
\end{aligned}$$

By Clairaut's theorem, assuming continuity of the mixed second partial derivatives, the differentiation is agnostic of order:

$$\frac{\partial^2 v_i}{\partial j \partial k} = \frac{\partial^2 v_i}{\partial k \partial j}$$

where the three colored pairs (after distribution) cancel pursuant to Clairaut's theorem. □

Using $\boldsymbol{v} = \boldsymbol{v}_a$, we have

$$\begin{aligned}
\nabla \times \boldsymbol{v}_a &= \begin{vmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 & 3xz^2 & -2xz \end{vmatrix} \\
&= -6xz\hat{\boldsymbol{x}} + 2z\hat{\boldsymbol{y}} + 3z^2\hat{\boldsymbol{z}} \\
\nabla \cdot (\nabla \times \boldsymbol{v}_a) &= \frac{\partial}{\partial x}(-6xz) + \frac{\partial}{\partial y}(2z) + \frac{\partial}{\partial z}\left(3z^2\right) \\
&= 0
\end{aligned}$$

Problem: 1.28

Prove that the curl of a gradient is always zero. *Check* it for function (b) in Prob. 1.11.

$\nabla \times (\nabla T) = \mathbf{0}.$

PROOF.

$$
\begin{aligned}
\nabla \times (\nabla T) &= \begin{vmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial T}{\partial x} & \frac{\partial T}{\partial y} & \frac{\partial T}{\partial z} \end{vmatrix} \\
&= \left(\frac{\partial^2 T}{\partial y \partial z} - \frac{\partial^2 T}{\partial z \partial y} \right) \hat{\boldsymbol{x}} + \left(\frac{\partial^2 T}{\partial z \partial x} - \frac{\partial^2 T}{\partial x \partial z} \right) \hat{\boldsymbol{y}} + \left(\frac{\partial^2 T}{\partial x \partial y} - \frac{\partial^2 T}{\partial y \partial x} \right) \hat{\boldsymbol{z}} \\
&= \mathbf{0}
\end{aligned}
$$

where the last equality follows from Clairaut's theorem. □

Using $f(x, y, z) = x^2 y^3 z^4$, calculate

$$
\begin{aligned}
\nabla f &= 2xy^3z^4 \hat{\boldsymbol{x}} + 3x^2y^2z^4 \hat{\boldsymbol{y}} + 4x^2y^3z^3 \hat{\boldsymbol{z}} \\
\nabla \times (\nabla f) &= \begin{vmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy^3z^4 & 3x^2y^2z^4 & 4x^2y^3z^3 \end{vmatrix} \\
&= \left[\frac{\partial}{\partial y} \left(4x^2y^3z^3\right) - \frac{\partial}{\partial z} \left(3x^2y^2z^4\right) \right] \hat{\boldsymbol{x}} \\
&+ \quad \left[\frac{\partial}{\partial z} \left(2xy^3z^4\right) - \frac{\partial}{\partial x} \left(4x^2y^3z^3\right) \right] \hat{\boldsymbol{y}} \\
&\quad + \left[\frac{\partial}{\partial x} \left(3x^2y^2z^4\right) - \frac{\partial}{\partial y} \left(2xy^3z^4\right) \right] \hat{\boldsymbol{z}} \\
&= \mathbf{0}
\end{aligned}
$$

1.3 - Integral Calculus

1.3.1 - Line, Surface, and Volume Integrals

Problem: 1.29

Calculate the line integral of the function $\boldsymbol{v} = x^2\hat{\boldsymbol{x}} + 2yz\hat{\boldsymbol{y}} + y^2\hat{\boldsymbol{z}}$ from the origin to the point $(1, 1, 1)$ by three different routes:

(a) $(0, 0, 0) \to (1, 0, 0) \to (1, 1, 0) \to (1, 1, 1)$.

(b) $(0, 0, 0) \to (0, 0, 1) \to (0, 1, 1) \to (1, 1, 1)$.

(c) The direct straight line.

(d) What is the line integral around the closed loop that goes *out* along path (a) and *back* along path (b)?

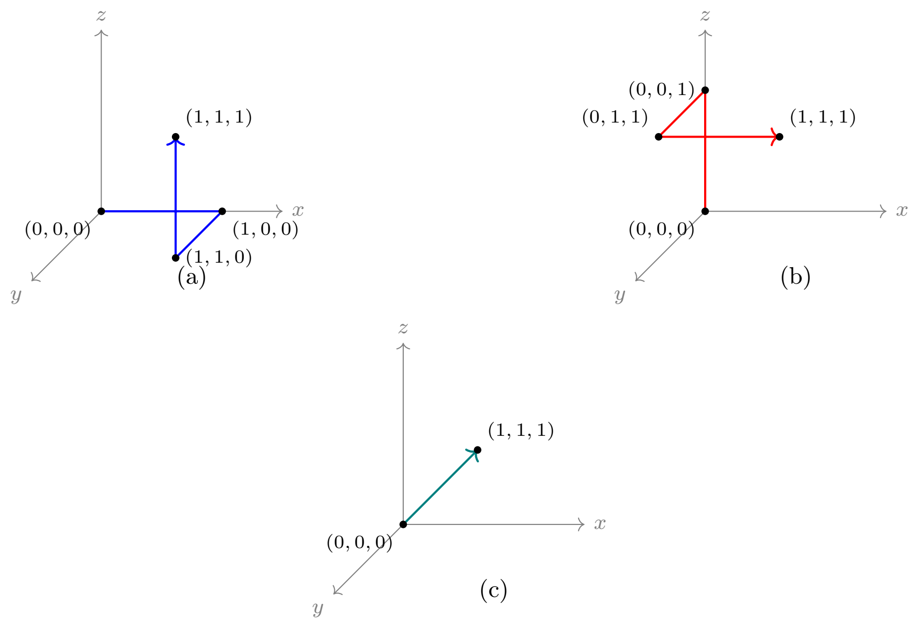


Figure 1: The three integration routes of Problem 1.29.

For all parts, $d\boldsymbol{l} = dx\hat{\boldsymbol{x}} + dy\hat{\boldsymbol{y}} + dz\hat{\boldsymbol{z}}$.

(a) $\underline{(0, 0, 0) \to (1, 0, 0)}$: We have $dy = dz = 0$ and $y = z = 0$. Calculate

$$\int_0^1 x^2\,dx = \frac{1}{3}$$

$\underline{(1, 0, 0) \to (1, 1, 0)}$: We have $dx = dz = 0$, $x = 1$, and $z = 0$. Then $\boldsymbol{v}\cdot d\boldsymbol{l} = 0$, leaving us with 0 contribution on this section.

$\underline{(1,1,0) \to (1,1,1)}$: We have $dx = dy = 0$ and $x = y = 1$. Calculate

$$\int_0^1 dz = 1$$

Adding the contributions from each path produces 4/3.

(b) $\underline{(0,0,0) \to (0,0,1)}$: We have $dx = dy = 0$ and $x = y = 0$. Since $\boldsymbol{v} \cdot d\boldsymbol{l} = 0$, the contribution on this section is 0.

$\underline{(0,0,1) \to (0,1,1)}$: We have $dx = dz = 0$, $x = 0$, and $z = 1$. Calculate

$$\int_0^1 2y\, dy = 1$$

$\underline{(0,1,1) \to (1,1,1)}$: We have $dy = dz = 0$ and $y = z = 1$. Calculate

$$\int_0^1 x^2\, dx = \frac{1}{3}$$

Summing each section's contribution, we have 4/3 yet again.

(c) Since the line connecting $(0,0,0)$ and $(1,1,1)$ is the set of all (x,y,z) such that $x = y = z$, parameterize as t. Then $dx = dy = dz = dt$, $d\boldsymbol{l} = dt(\hat{\boldsymbol{x}} + \hat{\boldsymbol{y}} + \hat{\boldsymbol{z}})$, and $\boldsymbol{v} = t^2\hat{\boldsymbol{x}} + 2t^2\hat{\boldsymbol{y}} + t^2\hat{\boldsymbol{z}}$. Thus $\boldsymbol{v} \cdot d\boldsymbol{l}$ gives us

$$\boldsymbol{v} \cdot d\boldsymbol{l} = 4t^2 dt$$

Integrating over $t \in [0,1]$, compute

$$\int_0^1 4t^2\, dt = \frac{4}{3}$$

(d)

$$\oint \boldsymbol{v} \cdot d\boldsymbol{l} = \frac{4}{3} - \frac{4}{3} = 0$$

These answers are no coincidence, for later we will learn that a vector field $\boldsymbol{v}$ with the property

$$\nabla \times \boldsymbol{v} = \mathbf{0}$$

is a **conservative** force. As it turns out, one can verify that this is the case for our $\boldsymbol{v}$.

Problem: 1.30

Calculate the surface integral of the function in Ex. 1.7, over the *bottom* of the box. For consistency, let "upward" be the positive direction. Does the surface integral depend only on the boundary line for this function? What is the total flux over the *closed* surface of the box (*including* the bottom)? [*Note*: For the *closed* surface, the positive direction is "outward," and hence "down," for the bottom face.]

We have $z = 0$, $d\boldsymbol{a} = -dx\,dy\hat{\boldsymbol{z}}$, and $\boldsymbol{v} \cdot d\boldsymbol{a} = 3y\,dx\,dy$, so we compute

$$\int_0^2 dx \int_0^2 3y\,dy = 12$$

From the flux of the other five sides, we sum: $20 + 12 = 32$. In this case, the surface integral depends on the surface itself. The boundary line in equation is the square spanned by $(0, 0, 0)$ and $(2, 2, 0)$, which is the boundary line shared by the bottom face, or by the total surface of the *other* five faces combined. Since each surface has a different flux, the surface integral is surface-dependent.

Problem: 1.31

Calculate the volume integral of the function $T = z^2$ over the tetrahedron with corners at $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$.

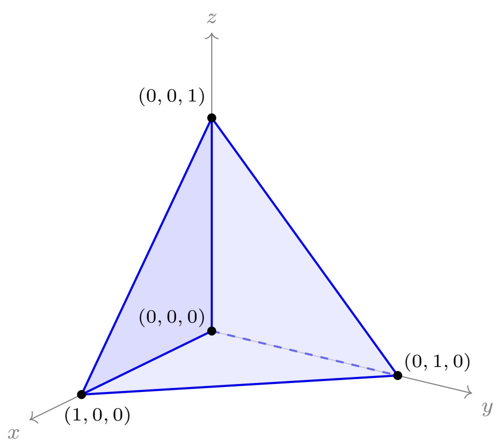


Figure 2: The tetrahedron of Problem 1.31, with vertices at the origin and the three unit axis points.

The plane traced out by the tetrahedron has shape $x + y + z = 1$. A clever approach is to consider a horizontal cross-section of the tetrahedron (i.e. fixing z at some constant), and the dimensions of the triangle traced out by that level surface:

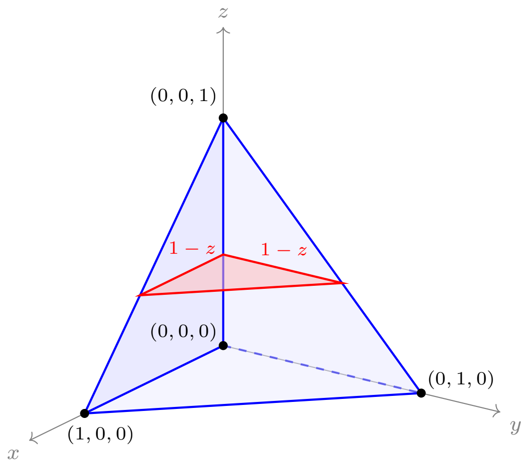


Figure 3: Horizontal cross-section of the tetrahedron at height z: a right triangle with legs $1-z$ and area $\frac{1}{2}(1-z)^2$.

Upon fixing the height of that level surface, we can see that the right triangle traced out by the level surface intersecting with the slanted face of the tetrahedron has legs of equal length $1-z$. The area of this triangle is $\frac{1}{2}(1-z)^2$; critically, we ought to make the observation that

$$\frac{1}{2}(1-z)^2 = dx\,dy$$

This simplifies our volume integral to

$$\int_0^1 z^2 \cdot \frac{1}{2}(1-z)^2 \, dz = \frac{1}{2}\int_0^1 \left(z^2 - 2z^3 + z^4\right) dz = \frac{1}{2}\left(\frac{1}{3} - \frac{1}{2} + \frac{1}{5}\right) = \frac{1}{60}$$

1.3.3 - The Fundamental Theorem of Gradients

Problem: 1.32

Check the fundamental theorem for gradients, using $T = x^2 + 4xy + 2yz^3$, the points $\boldsymbol{a} = (0,0,0)$, $\boldsymbol{b} = (1,1,1)$, and the three paths in Fig. 1.28:

(a) $(0,0,0) \to (1,0,0) \to (1,1,0) \to (1,1,1)$;

(b) $(0,0,0) \to (0,0,1) \to (0,1,1) \to (1,1,1)$;

(c) the parabolic path $z = x^2$; $y = x$.

The gradient is

$$\nabla T = (2x+4y)\,\hat{\boldsymbol{x}} + \left(4x + 2z^3\right)\hat{\boldsymbol{y}} + 6yz^2\,\hat{\boldsymbol{z}}$$

(a) (i) $\underline{(0,0,0) \to (1,0,0)}$:

$$\begin{aligned} y = z = dy = dz &= 0 \\ \nabla T \cdot d\boldsymbol{l} &= 2x\,dx \end{aligned}$$

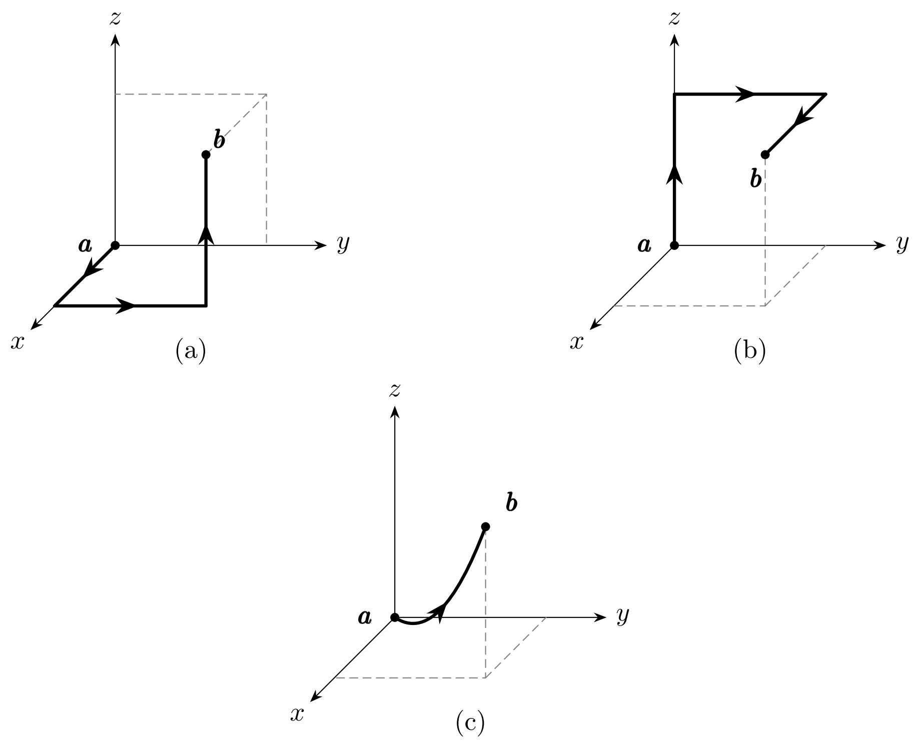


Figure 4: The three paths of Problem 1.32 (cf. Griffiths Fig. 1.28).

$$\int_0^1 2x\,dx = 1$$

(ii) $\underline{(1,0,0) \rightarrow (1,1,0)}$:

$$x = 1$$
$$z = dx = dz = 0$$
$$\nabla T \cdot d\boldsymbol{l} = \left(4x + 2z^3\right)\,dy$$

$$\int_0^1 4\,dy = 4$$

(iii) $\underline{(1,1,0) \rightarrow (1,1,1)}$:

$$x = y = 1$$
$$dx = dy = 0$$
$$\nabla T \cdot d\boldsymbol{l} = 6z^2\,dz$$

$$\int_0^1 6z^2\,dz = 2$$

Combining the path contributions,

$$\int_{\boldsymbol{a}}^{\boldsymbol{b}} \nabla T \cdot d\boldsymbol{l} = 7$$

(b) (i) $\underline{(0,0,0) \to (0,0,1)}$:

$$\begin{aligned} x = y = dx = dy &= 0 \\ \nabla T \cdot d\boldsymbol{l} &= 0 \\ \int_0^1 0 \, dz &= 0 \end{aligned}$$

(ii) $\underline{(0,0,1) \to (0,1,1)}$:

$$\begin{aligned} x = dx = dz &= 0 \\ z &= 1 \\ \nabla T \cdot d\boldsymbol{l} &= 2\, dy \\ \int_0^1 2\, dy &= 2 \end{aligned}$$

(iii) $\underline{(0,1,1) \to (1,1,1)}$:

$$\begin{aligned} y = z &= 1 \\ dy = dz &= 0 \\ \nabla T \cdot d\boldsymbol{l} &= (2x + 4)\ dx \\ \int_0^1 (2x+4)\ dx &= 5 \end{aligned}$$

Combining the path contributions, we derive the same result, as anticipated from the fundamental theorem for gradients.

$$\int_{\boldsymbol{a}}^{\boldsymbol{b}} \nabla T \cdot d\boldsymbol{l} = 7$$

(c) From the parabola definition, derive

$$dz = 2x\, dx, \qquad dy = dx$$

Then the infinitesimal displacement becomes

$$d\boldsymbol{l} = (\hat{\boldsymbol{x}} + \hat{\boldsymbol{y}} + 2x\hat{\boldsymbol{z}})\, dx$$

and the gradient, now written exclusively in x:

$$\nabla T = 6x\hat{\boldsymbol{x}} + \left(4x + 2x^6\right) \hat{\boldsymbol{y}} + 6x^5 \hat{\boldsymbol{z}}$$

Compute the dot product as

$$\nabla T \cdot d\boldsymbol{l} = 10x + 14x^6$$

Integrating the line integral, compute

$$\int_0^1 \left(10x + 14x^6\right)\ dx = \left(5x^2 + 2x^7\right)\Big|_0^1 = 7$$

As expected, the fundamental theorem of gradient works!

1.3.4 - The Fundamental Theorem for Divergences

Problem: 1.33

Test the divergence theorem for the function $\boldsymbol{v} = (xy)\,\hat{\boldsymbol{x}} + (2yz)\,\hat{\boldsymbol{y}} + (3zx)\,\hat{\boldsymbol{z}}$. Take as your volume the cube shown in Fig. 1.30, with sides of length 2.

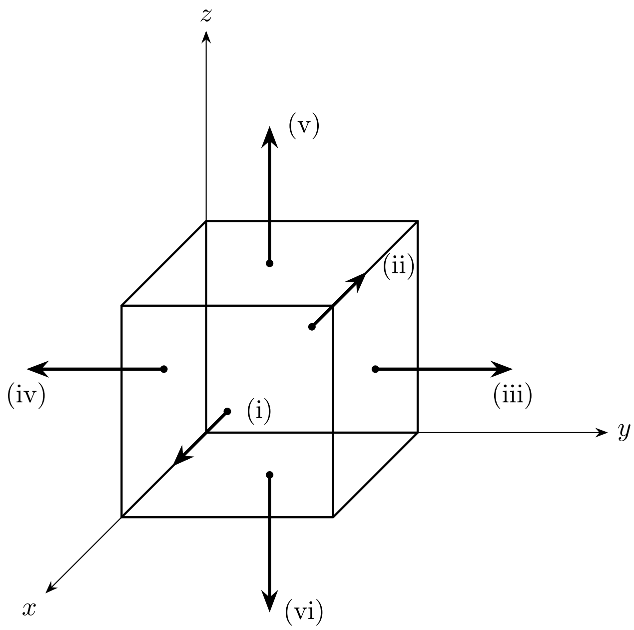


Figure 5: Cube of length 2 with outward normals on each face (cf. Griffiths Fig. 1.30).

The divergence is

$$\nabla \cdot \boldsymbol{v} = 3x + y + 2z$$

Then its volume integral is

$$\int_{\mathcal{V}} \nabla \cdot \boldsymbol{v}\, d\tau = \int_0^2 \int_0^2 \int_0^2 (3x + y + 2z)\; dx\, dy\, dz = 48$$

Now for the surface integral:

$$\oint_{\mathcal{S}} \boldsymbol{v} \cdot d\boldsymbol{a}$$

(i) We have $x = 2$ and $\boldsymbol{v} \cdot dy\, dz\hat{\boldsymbol{x}} = 2y\, dy\, dz$.

$$\int \boldsymbol{v} \cdot d\boldsymbol{a} = \int_0^2 \int_0^2 2y\, dy\, dz = 8$$

(ii) $x = 0$ and $-\boldsymbol{v} \cdot dy\, dz\hat{\boldsymbol{x}} = 0\, dy\, dz$.

$$\int \boldsymbol{v} \cdot d\boldsymbol{a} = \int_0^2 \int_0^2 0\, dy\, dz = 0$$

(iii) $y = 2$ and $\boldsymbol{v} \cdot dx\, dz\hat{\boldsymbol{y}} = 4z\, dx\, dz$.

$$\int \boldsymbol{v} \cdot d\boldsymbol{a} = \int_0^2 \int_0^2 4z\, dx\, dz = 16$$

(iv) $y = 0$ and $-\boldsymbol{v} \cdot dx\, dz\hat{\boldsymbol{y}} = 0\, dx\, dz$.

$$\int \boldsymbol{v} \cdot d\boldsymbol{a} = \int_0^2 \int_0^2 0\, dx\, dz = 0$$

(v) $z = 2$ and $\boldsymbol{v} \cdot dx\, dy\hat{\boldsymbol{z}} = 6x\, dx\, dy$.

$$\int \boldsymbol{v} \cdot d\boldsymbol{a} = \int_0^2 \int_0^2 6x\, dx\, dy = 24$$

(vi) $z = 0$ and $-\boldsymbol{v} \cdot dx\, dy\hat{\boldsymbol{z}} = 0\, dx\, dy$.

$$\int \boldsymbol{v} \cdot d\boldsymbol{a} = \int_0^2 \int_0^2 0\, dx\, dy = 0$$

Summing the fluxes produces

$$\oint_{\mathcal{S}} \boldsymbol{v} \cdot d\boldsymbol{a} = 8 + 16 + 24 = 48$$

verifying the divergence theorem.

1.3.5 - The Fundamental Theorem for Curls

Problem: 1.34

Test Stokes' theorem for the function $\boldsymbol{v} = (xy)\,\hat{\boldsymbol{x}} + (2yz)\,\hat{\boldsymbol{y}} + (3zx)\,\hat{\boldsymbol{z}}$ using the triangular shaded area of Fig. 1.34.

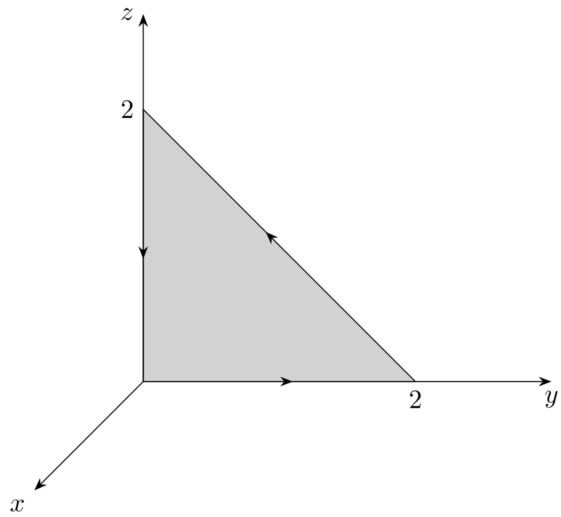


Figure 6: Griffiths Figure 1.34

Compute

$$\nabla \times \boldsymbol{v} = \begin{vmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy & 2yz & 3zx \end{vmatrix}$$
$$= -2y\hat{\boldsymbol{x}} - 3z\hat{\boldsymbol{y}} - x\hat{\boldsymbol{z}}$$

The surface integral is

$$\begin{aligned} \int_{\mathcal{S}} (\nabla \times \boldsymbol{v}) \cdot d\boldsymbol{a} &= -\int_0^2 \int_0^{2-z} 2y\,dy\,dz \\ &= -\int_0^2 (2-z)^2 \; dz \\ &= -4z + 2z^2 - \frac{z^3}{3}\Big|_0^2 \\ &= -\frac{8}{3} \end{aligned}$$

Now for the line integral:

(i) $\underline{(0,0,0) \to (0,2,0)}$: $x = z = dx = dz = 0$, $d\boldsymbol{l} = dy\hat{\boldsymbol{y}}$, $\boldsymbol{v} \cdot d\boldsymbol{l} = 0\,dy$

$$\int \boldsymbol{v} \cdot d\boldsymbol{l} = 0$$

(ii) $\underline{(0,2,0) \to (0,0,2)}$:

$$\begin{aligned} x = dx &= 0 \\ d\boldsymbol{l} &= dy\hat{\boldsymbol{y}} + dz\hat{\boldsymbol{z}} = (\hat{\boldsymbol{y}} - \hat{\boldsymbol{z}})\,dy \\ \boldsymbol{v}\cdot d\boldsymbol{l} &= 2yz\,dy = 2y(2-y)\;dy \end{aligned}$$

where the hypotenuse of the triangle has equation $z = 2 - y$, producing $dz/dy = -1 \iff dz = -dy$. This segment's contribution is

$$\int \boldsymbol{v}\cdot d\boldsymbol{l} = \int_2^0 2y(2-y)\;dy = \frac{2}{3}y^3 - 2y^2\Big|_0^2 = -\frac{8}{3}$$

where the bounds are flipped to reflect the movement along the hypotenuse in the negative y-direction.

(iii) $\underline{(0,0,2) \to (0,0,0)}$: $x = y = dx = dy = 0$, $d\boldsymbol{l} = dz\hat{\boldsymbol{z}}$, $\boldsymbol{v}\cdot d\boldsymbol{l} = 0\,dz$

$$\int \boldsymbol{v}\cdot d\boldsymbol{l} = 0$$

Summed together, we verify Stokes' theorem:

$$\int (\nabla \times \boldsymbol{v})\cdot d\boldsymbol{a} = \oint \boldsymbol{v}\cdot d\boldsymbol{l} = -\frac{8}{3}$$

Problem: 1.35

Check Corollary 1 by using the same function and boundary line as in Ex. 1.11, but integrating over the five faces of the cube in Fig. 1.35. The back of the cube is open.

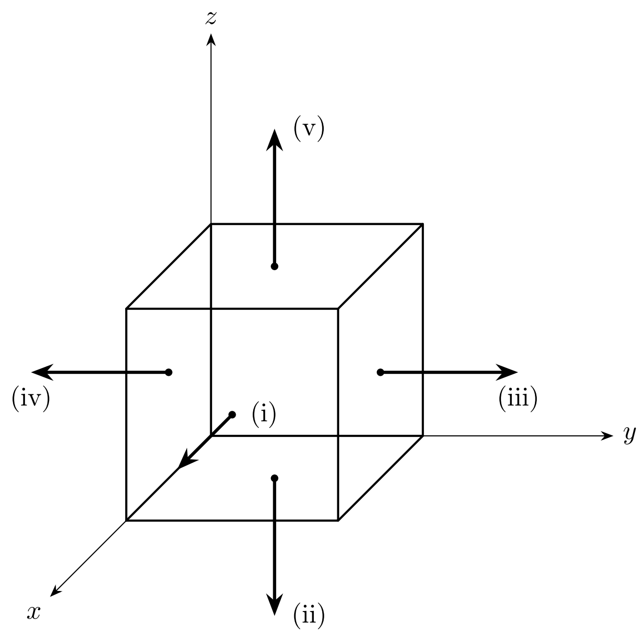


Figure 7: Unit cube with outward normals on each face (cf. Griffiths Fig. 1.35).

Our function is $\boldsymbol{v} = \left(2xz + 3y^2\right)\hat{\boldsymbol{y}} + \left(4yz^2\right)\hat{\boldsymbol{z}}$, with

$$\nabla \times \boldsymbol{v} = \left(4z^2 - 2x\right)\hat{\boldsymbol{x}} + 2z\hat{\boldsymbol{z}}$$

computed in Ex. 1.11. Corollary 1 states that $\int_{\mathcal{S}} (\nabla \times \boldsymbol{v}) \cdot d\boldsymbol{a}$ depends only on the boundary line, so integrating over the five faces should reproduce the value $\frac{4}{3}$ obtained in Ex. 1.11 for the square surface at $x = 0$. The right-hand rule applied to the boundary circulation makes the normals point outward.

(i) $x = 1$, $d\boldsymbol{a} = dy\,dz\hat{\boldsymbol{x}}$, $(\nabla \times \boldsymbol{v}) \cdot d\boldsymbol{a} = \left(4z^2 - 2\right)\,dy\,dz$.

$$\int (\nabla \times \boldsymbol{v}) \cdot d\boldsymbol{a} = \int_0^1 \int_0^1 \left(4z^2 - 2\right)\,dy\,dz = \frac{4}{3} - 2 = -\frac{2}{3}$$

(ii) $z = 0$, $d\boldsymbol{a} = -dx\,dy\hat{\boldsymbol{z}}$, $(\nabla \times \boldsymbol{v}) \cdot d\boldsymbol{a} = -2z\,dx\,dy = 0\,dx\,dy$.

$$\int (\nabla \times \boldsymbol{v}) \cdot d\boldsymbol{a} = 0$$

(iii) $y = 1$, $d\boldsymbol{a} = dx\,dz\hat{\boldsymbol{y}}$, and $\nabla \times \boldsymbol{v}$ has no $\hat{\boldsymbol{y}}$ component.

$$\int (\nabla \times \boldsymbol{v}) \cdot d\boldsymbol{a} = 0$$

(iv) $y = 0$, $d\boldsymbol{a} = -dx\,dz\hat{\boldsymbol{y}}$, likewise.

$$\int (\nabla \times \boldsymbol{v}) \cdot d\boldsymbol{a} = 0$$

(v) $z = 1$, $d\boldsymbol{a} = dx\,dy\hat{\boldsymbol{z}}$, $(\nabla \times \boldsymbol{v}) \cdot d\boldsymbol{a} = 2z\,dx\,dy = 2\,dx\,dy$.

$$\int (\nabla \times \boldsymbol{v}) \cdot d\boldsymbol{a} = \int_0^1 \int_0^1 2\,dx\,dy = 2$$

Summing the fluxes produces

$$\int_{\mathcal{S}} (\nabla \times \boldsymbol{v}) \cdot d\boldsymbol{a} = -\frac{2}{3} + 2 = \frac{4}{3}$$

matching Ex. 1.11 and confirming Corollary 1.

1.3.6 - Integration by Parts

Problem: 1.36

(a) Show that

$$\int_{\mathcal{S}} f(\nabla \times \boldsymbol{A}) \cdot d\boldsymbol{a} = \int_{\mathcal{S}} [\boldsymbol{A} \times (\nabla f)] \cdot d\boldsymbol{a} + \oint_{\mathcal{P}} f\boldsymbol{A} \cdot d\boldsymbol{l}$$

(b) Show that

$$\int_{\mathcal{V}} \boldsymbol{B} \cdot (\nabla \times \boldsymbol{A})\ d\tau = \int_{\mathcal{V}} \boldsymbol{A} \cdot (\nabla \times \boldsymbol{B})\ d\tau + \oint_{\mathcal{S}} (\boldsymbol{A} \times \boldsymbol{B}) \cdot d\boldsymbol{a}$$

(a) From the product rule

$$\nabla \times (f\boldsymbol{A}) = f(\nabla \times \boldsymbol{A}) - \boldsymbol{A} \times (\nabla f)$$

Rewrite as

$$f(\nabla \times \boldsymbol{A}) = \boldsymbol{A} \times (\nabla f) + \nabla \times (f\boldsymbol{A})$$

Take the surface integral of each term to find

$$\int_{\mathcal{S}} f(\nabla \times \boldsymbol{A}) \cdot d\boldsymbol{a} = \int_{\mathcal{S}} [\boldsymbol{A} \times (\nabla f)] \cdot d\boldsymbol{a} + \int_{\mathcal{S}} \nabla \times (f\boldsymbol{A}) \cdot d\boldsymbol{a}$$

On the last term, apply Stokes' theorem to convert to a loop integral, leaving us with

$$\int_{\mathcal{S}} f(\nabla \times \boldsymbol{A}) \cdot d\boldsymbol{a} = \int_{\mathcal{S}} [\boldsymbol{A} \times (\nabla f)] \cdot d\boldsymbol{a} + \oint_{\mathcal{P}} f\boldsymbol{A} \cdot d\boldsymbol{l}$$

(b) From the product rule

$$\nabla \cdot (\boldsymbol{A} \times \boldsymbol{B}) = \boldsymbol{B} \cdot (\nabla \times \boldsymbol{A}) - \boldsymbol{A} \cdot (\nabla \times \boldsymbol{B})$$

Rewrite as

$$\boldsymbol{B} \cdot (\nabla \times \boldsymbol{A}) = \boldsymbol{A} \cdot (\nabla \times \boldsymbol{B}) + \nabla \cdot (\boldsymbol{A} \times \boldsymbol{B})$$

Take the volume integral of each term to find

$$\int_{\mathcal{V}} \boldsymbol{B} \cdot (\nabla \times \boldsymbol{A})\ d\tau = \int_{\mathcal{V}} \boldsymbol{A} \cdot (\nabla \times \boldsymbol{B})\ d\tau + \int_{\mathcal{V}} \nabla \cdot (\boldsymbol{A} \times \boldsymbol{B})\ d\tau$$

By the divergence theorem, convert the last term to the surface flux to derive

$$\int_{\mathcal{V}} \boldsymbol{B} \cdot (\nabla \times \boldsymbol{A})\ d\tau = \int_{\mathcal{V}} \boldsymbol{A} \cdot (\nabla \times \boldsymbol{B})\ d\tau + \oint_{\mathcal{S}} (\boldsymbol{A} \times \boldsymbol{B}) \cdot d\boldsymbol{a}$$

1.4 - Curvilinear Coordinates

Problem: 1.37

Find formulas for r, θ, ϕ in terms of x, y, z (the inverse, in other words, of Eq. 1.62).

The transformation from Cartesian to spherical coordinates is given by Equation 1.62:

$$x = r\sin(\theta)\cos(\phi)\,, \qquad y = r\sin(\theta)\sin(\phi)\,, \qquad z = r\cos(\theta)$$

The radial length is derived as the positive root of

$$\begin{aligned} r^2 &= r^2\left[\sin^2(\theta) + \cos^2(\theta)\right] \\ &= \left[r^2\sin^2(\theta)\cos^2(\phi) + r^2\sin^2(\theta)\sin^2(\phi)\right] + r^2\cos^2(\theta) \\ &= x^2 + y^2 + z^2 \end{aligned}$$

For θ, in the z-equation, isolate θ:

$$\frac{z}{r} = \frac{z}{\sqrt{x^2+y^2+z^2}} = \cos(\theta) \qquad \Longrightarrow \qquad \theta = \cos^{-1}\left(\frac{z}{\sqrt{x^2+y^2+z^2}}\right)$$

Lastly, for ϕ, divide y by x to find

$$\frac{y}{x} = \tan(\phi) \qquad \Longrightarrow \qquad \phi = \tan^{-1}\left(\frac{y}{x}\right)$$

which is specifically defined on $(-\pi/2, \pi/2)$.

Problem: 1.38

Express the unit vectors $\hat{\boldsymbol{r}}$, $\hat{\boldsymbol{\theta}}$, $\hat{\boldsymbol{\phi}}$ in terms of $\hat{\boldsymbol{x}}$, $\hat{\boldsymbol{y}}$, $\hat{\boldsymbol{z}}$ (that is, derive Eq. 1.64). Check your answers several ways ($\hat{\boldsymbol{r}}\cdot\hat{\boldsymbol{r}} \stackrel{?}{=} 1$, $\hat{\boldsymbol{\theta}}\cdot\hat{\boldsymbol{\phi}} \stackrel{?}{=} 0$, $\hat{\boldsymbol{r}}\times\hat{\boldsymbol{\theta}} \stackrel{?}{=} \hat{\boldsymbol{\phi}}$...). Also work out the inverse formulas, giving $\hat{\boldsymbol{x}}$, $\hat{\boldsymbol{y}}$, $\hat{\boldsymbol{z}}$ in terms of $\hat{\boldsymbol{r}}$, $\hat{\boldsymbol{\theta}}$, $\hat{\boldsymbol{\phi}}$ (and θ, ϕ).

Using the mappings from (x, y, z) to (r, θ, ϕ) in Equation 1.62, we can first derive our unit radial vector as

$$\hat{\boldsymbol{r}} = \frac{\boldsymbol{r}}{|\boldsymbol{r}|} = \sin(\theta)\cos(\phi)\,\hat{\boldsymbol{x}} + \sin(\theta)\sin(\phi)\,\hat{\boldsymbol{y}} + \cos(\theta)\,\hat{\boldsymbol{z}}$$

where $|\boldsymbol{r}| = r$. To find the direction of increasing θ, which is definitionally what $\hat{\boldsymbol{\theta}}$ is, we first differentiate $\hat{\boldsymbol{r}}$ with respect to θ to get $\boldsymbol{\theta} = \partial\hat{\boldsymbol{r}}/\partial\theta$, and normalize to find

$$\hat{\boldsymbol{\theta}} = \frac{\partial\hat{\boldsymbol{r}}/\partial\theta}{|\partial\hat{\boldsymbol{r}}/\partial\theta|} = \frac{\boldsymbol{\theta}}{|\boldsymbol{\theta}|} = \cos(\theta)\cos(\phi)\,\hat{\boldsymbol{x}} + \cos(\theta)\sin(\phi)\,\hat{\boldsymbol{y}} - \sin(\theta)\,\hat{\boldsymbol{z}}$$

where $|\boldsymbol{\theta}| = 1$. Lastly, differentiate $\hat{\boldsymbol{r}}$ with respect to ϕ to get $\boldsymbol{\phi} = \partial\hat{\boldsymbol{r}}/\partial\phi$, and normalize:

$$\hat{\boldsymbol{\phi}} = \frac{\partial\hat{\boldsymbol{r}}/\partial\phi}{|\partial\hat{\boldsymbol{r}}/\partial\phi|} = \frac{\boldsymbol{\phi}}{|\boldsymbol{\phi}|} = -\sin(\phi)\,\hat{\boldsymbol{x}} + \cos(\phi)\,\hat{\boldsymbol{y}}$$

where $|\boldsymbol{\phi}| = \sin(\theta)$. Verifying in the recommended ways, we can compute

$$\begin{aligned}
\hat{\boldsymbol{r}} \cdot \hat{\boldsymbol{r}} &= \sin^2(\theta)\left[\cos^2(\phi) + \sin^2(\phi)\right] + \cos^2(\theta) \\
&= \sin^2(\theta) + \cos^2(\theta) \\
&= 1 \\
\hat{\boldsymbol{\theta}} \cdot \hat{\boldsymbol{\phi}} &= -\cos(\theta)\cos(\phi)\sin(\phi) + \cos(\theta)\sin(\phi)\cos(\phi) \\
&= 0 \\
\hat{\boldsymbol{r}} \times \hat{\boldsymbol{\theta}} &= \begin{vmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ \sin(\theta)\cos(\phi) & \sin(\theta)\sin(\phi) & \cos(\theta) \\ \cos(\theta)\cos(\phi) & \cos(\theta)\sin(\phi) & -\sin(\theta) \end{vmatrix} \\
&= -\sin(\phi)\,\hat{\boldsymbol{x}} + \cos(\phi)\,\hat{\boldsymbol{y}} \\
&= \hat{\boldsymbol{\phi}}
\end{aligned}$$

Finally, the inverse formulas. We may write our existing formulas as follows:

$$\begin{pmatrix} \sin(\theta)\cos(\phi) & \sin(\theta)\sin(\phi) & \cos(\theta) \\ \cos(\theta)\cos(\phi) & \cos(\theta)\sin(\phi) & -\sin(\theta) \\ -\sin(\phi) & \cos(\phi) & 0 \end{pmatrix} \begin{pmatrix} \hat{\boldsymbol{x}} \\ \hat{\boldsymbol{y}} \\ \hat{\boldsymbol{z}} \end{pmatrix} = \begin{pmatrix} \hat{\boldsymbol{r}} \\ \hat{\boldsymbol{\theta}} \\ \hat{\boldsymbol{\phi}} \end{pmatrix}$$

Now, take note of this insight: the rows of the coefficient matrix form an **orthonormal basis**, rendering the matrix itself **orthogonal**. Therefore it satisfies the property

$$A^{-1} = A^{\mathrm{T}}$$

Ergo, we can immediately write

$$\begin{pmatrix} \hat{\boldsymbol{x}} \\ \hat{\boldsymbol{y}} \\ \hat{\boldsymbol{z}} \end{pmatrix} = \begin{pmatrix} \sin(\theta)\cos(\phi) & \cos(\theta)\cos(\phi) & -\sin(\phi) \\ \sin(\theta)\sin(\phi) & \cos(\theta)\sin(\phi) & \cos(\phi) \\ \cos(\theta) & -\sin(\theta) & 0 \end{pmatrix} \begin{pmatrix} \hat{\boldsymbol{r}} \\ \hat{\boldsymbol{\theta}} \\ \hat{\boldsymbol{\phi}} \end{pmatrix}$$

which leaves us with

$$\begin{aligned}
\hat{\boldsymbol{x}} &= \sin(\theta)\cos(\phi)\,\hat{\boldsymbol{r}} + \cos(\theta)\cos(\phi)\,\hat{\boldsymbol{\theta}} - \sin(\phi)\,\hat{\boldsymbol{\phi}} \\
\hat{\boldsymbol{y}} &= \sin(\theta)\sin(\phi)\,\hat{\boldsymbol{r}} + \cos(\theta)\sin(\phi)\,\hat{\boldsymbol{\theta}} + \cos(\phi)\,\hat{\boldsymbol{\phi}} \\
\hat{\boldsymbol{z}} &= \cos(\theta)\,\hat{\boldsymbol{r}} - \sin(\theta)\,\hat{\boldsymbol{\theta}}
\end{aligned}$$

Problem: 1.39

(a) Check the divergence theorem for the function $\boldsymbol{v}_1 = r^2\hat{\boldsymbol{r}}$, using as your volume the sphere of radius R, centered at the origin.

(b) Do the same for $\boldsymbol{v}_2 = \left(1/r^2\right)\hat{\boldsymbol{r}}$. (If the answer surprises you, look back at Prob. 1.16.)

First calculate the integrand

$$\begin{aligned}
\nabla \cdot \boldsymbol{v}_1 &= \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^4\right) \\
&= 4r
\end{aligned}$$

which we evaluate as

$$\int_{\mathcal{V}}$$